



**British
Antarctic Survey**
NATURAL ENVIRONMENT RESEARCH COUNCIL

Development of a Prototype Climate-monitoring Sensor Buoy for Polar Regions

MEng Project Team 4

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List of Abbreviations

ADC	Analogue to Digital Converter
ALS	Ambient Light Sensing
ATT MTU	Attribute Protocol Maximum Transmission Unit
AUV	Autonomous Underwater Vehicle
BAS	British Antarctic Survey
BLE	Bluetooth Low Energy
BMS	Battery Management System
BRISQUE	Blind/Referenceless Image Spatial Quality Evaluator
CAD	Computer-Aided Design
CNC	Computer Numerical Control
CRC	Cyclic Redundancy Check
CRREL	Cold Regions Research and Engineering Laboratory
CSV	Comma-Separated Values
DAC	Digital to Analogue Converter
DC	Direct Current
DPR	Dark Pixel Ratio
DR	Design Requirement
ECV	Essential Climate Variable
EMI	Electromagnetic Induction
EN	Enable
ESA	European Space Agency
ETFE	Ethylene Tetrafluoroethylene
FFT	Fast Fourier Transform
FR	Full-Reference
GATT	Generic Attribute Profile
GCOS	Global Climate Observing System
GPS	Global Positioning System
GUI	Graphical User Interface
HCP	Horizontal Coplanar
HDPE	High-Density Polyethylene
HEM	Helicopter Electromagnetic
IC	Integrated Circuit
IEC	International Electrotechnical Commission
IMB	Ice Mass-Balance Buoy
IMU	Inertial Measurement Unit
INA	Instrumentation Amplifier
IPCC	Intergovernmental Panel on Climate Change
IQA	Image Quality Assessment
IR	Infrared
ISO	International Organisation for Standardisation
JPEG	Joint Photographic Experts Group
LED	Light-Emitting Diode
Li-Ion	Lithium-Ion
LiFePO ₄	Lithium Iron Phosphate
Li-SOCl ₂	Lithium-Thionyl Chloride
LTO	Lithium Titanate

MCU	Microcontroller Unit
MEMS	Micro-Electro-Mechanical System
MO	Mobile Originated
MPP	Maximum Power Point
MPPT	Maximum Power Point Tracking
MQS	Magnetic Quasi-Static
MT	Mobile Terminated
NATO	North Atlantic Treaty Organisation
NC	Normally Closed
NCA	Nickel Cobalt Aluminium
NIR	Near Infrared
NMC	Nickel Manganese Cobalt
NMEA	National Marine Electronics Association
NOAA	National Oceanic and Atmospheric Administration
NR	No-Reference
NTC	Negative Temperature Coefficient
NTU	Nephelometric Turbidity Units
OD	Outer Diameter
PCB	Printed Circuit Board
PLA	Polylactic Acid
PRT	Platinum Resistance Thermometer
PV	Photovoltaic
PWM	Pulse Width Modulation
QQVGA	Quarter Quarter Video Graphics Array
RMS	Root Mean Square
RTC	Real-Time Clock
RX	Receive / Receiver
SBD	Short Burst Data
SIMB	Seasonal Ice Mass-Balance Buoy
SIT	Sea Ice Thickness
SLA	Sealed Lead-Acid
SNR	Signal-to-Noise Ratio
SoC	State of Charge
SRAM	Static Random-Access Memory
STC	Standard Test Conditions
TIA	Transimpedance Amplifier
TSS	Total Suspended Solids
TX	Transmit / Transmitter
UART	Universal Asynchronous Receiver-Transmitter
ULS	Upward-Looking Sonar
UUID	Universally Unique Identifier
UV	Ultraviolet
UVI	Ultraviolet Index
UVS	Ultraviolet Sensor
VCC	Voltage at Common Collector (supply voltage)
VCP	Vertical Coplanar
VOC	Open-Circuit Voltage
WMO	World Meteorological Organisation

Executive Summary

The Arctic is warming nearly four times faster than the global average, yet accurate, continuous measurement of sea-ice thickness at meaningful spatial scales remains an unsolved challenge. Existing monitoring approaches, whether satellite-based, manual in-situ, or autonomous drifting buoys, each carry fundamental limitations in cost, spatial coverage, or operational dependency on human presence. This project is motivated by the critical need for an autonomous, scalable platform capable of measuring sea-ice thickness non-destructively alongside a broader suite of Essential Climate Variables, developed in direct collaboration with the British Antarctic Survey (BAS), whose researchers require precisely this capability to support their ongoing work in atmosphere-ice-ocean interaction.

The aim of this project has been to design, build, and test a prototype climate-monitoring sensor buoy suitable for autonomous operation in polar environments, capable of continuous data acquisition and communication over extended deployment periods without human intervention.

The objectives have comprised of: (1) a literature review of existing polar monitoring solutions; (2) design and implementation of a camera-based imaging system with onboard image quality filtering; (3) development of a modular environmental sensor suite; (4) design of custom microcontroller hardware to interface all subsystems; (5) development of a cold-resilient, energy-harvesting power architecture; (6) design and construction of a mechanical enclosure within the project specification; (7) integration of all hardware into a unified prototype; (8) implementation of a low data rate communications system; (9) development of a graphical user interface for data visualisation and control; and (10) full end-to-end system integration and testing.

A non-destructive frequency domain resonant EMI system was developed for sea ice thickness measurement, incorporating a bucking coil and discrete lock-in amplifier to separate the in-phase and quadrature components of the received signal. Additional signal conditioning in the form of phasor rotation and in-phase subtraction is applied to fully null the primary field contribution. The development process made use of CST simulations to characterise the electromagnetic environment and evaluate coil geometries prior to physical implementation. The system successfully demonstrated primary field nulling during laboratory testing, with a clear and consistent trend evident in the measured data across the range of target standoff distances evaluated. Further development is required to optimise the bucking coil geometry and to collect seawater target data, such that ice thickness can be accurately determined from the system's response.

A suite of environmental sensors was developed, tested, and validated, comprising both custom in-house designed solutions and selected commercial components. A custom turbidity sensor was designed and built around a 180-degree optical backscatter configuration, calibrated using kaolin clay and validated up to 400 NTU, demonstrating an exponential response. A custom conductivity-based salinity sensor was developed and validated, capable of measuring seawater conductivity up to 0.4 S/m, covering the full range of polar seawater conditions. External temperature measurement was implemented using a thermistor string mounted around the buoy body, achieving a resolution of $\pm 0.1^{\circ}\text{C}$, validated through ice-bath testing. UV Index measurement was implemented using the LTR-390UV-01 sensor, providing both UVI and ambient light readings. Wave spectra were characterised using the MPU6050 IMU mounted on the auxiliary daughter board, with a Fast Fourier Transform applied to extract the dominant wave frequency and a 16-bin power spectrum at a spectral resolution of 0.3125 Hz per bin.

A Bluetooth Low Energy (BLE) communications system was implemented, enabling remote data transmission and device configuration representative of satellite communication bandwidth constraints. A graphical user interface was developed to request, organise, and visualise data from all onboard sensors,

displaying readings on demand across 12 sensor channels. Custom microcontroller hardware comprising a central motherboard and daughter board was engineered to interface all subsystems. Unified hardware-firmware system integration was successfully executed, with all individual electronic and sensing subsystems verified to operate synchronously within design parameters.

A cold-resilient, energy-harvesting power architecture was developed to sustain multi-season autonomy, specifically designed to survive a 105-day winter darkness period at temperatures down to -40°C . Mathematical modelling validated a 2S12P Lithium-Ion energy storage, safeguarded by a hardware-enforced thermal cutoff that physically inhibits charging below 0°C to prevent lithium plating. For charging, a Maximum Power Point Tracking (MPPT) solar network was implemented, yielding 13.4 Wh/day under representative polar conditions. A custom charging and distribution PCB was fabricated, outputting four isolated voltage rails, including a gated $\pm 15\text{ V}$ supply for the EMI array. Transient load testing validated this architecture, demonstrating negligible voltage droop on the 3.3 V logic rail, proving system stability.

Future work is centred on transitioning the laboratory-validated prototype into a field-deployable platform. Mechanical priorities include fabricating a cold-resilient internal chassis from High-Density Polyethylene (HDPE), integrating an optically transparent outer hull, validating waterproofing, and designing an air deployment system for high-altitude mass deployment. Electrical development will focus on replacing prototype modules with discrete ICs, integrating the full-scope 2S12P battery architecture, and exploring active heating for sub-zero charging. Software and sensing enhancements will implement Iridium satellite telemetry, memory-efficient edge-processing for high-resolution imaging, and generative climate modelling. Together with environmental chamber testing, these steps will progress the system toward autonomous deployment in collaboration with the British Antarctic Survey.

1. Introduction

1.1. Background

The Arctic is warming nearly four times faster than the global average, a rate so far beyond model projections that it has forced scientists to fundamentally reassess the pace of polar climate change [1]. Central to this change is sea ice, which acts as a critical regulator of the global climate system. Its high reflectivity limits solar heat absorption at the surface, while its formation and melt cycles drive deep ocean circulation and govern heat exchange between the ocean and atmosphere [2]. As sea ice retreats, these mechanisms weaken, contributing to further warming in a self-reinforcing cycle. The Intergovernmental Panel on Climate Change (IPCC) has formally identified sea-ice thickness as an Essential Climate Variable (ECV) under the Global Climate Observing System (GCOS), recognising it as one of the parameters most critical to the understanding and prediction of climate change [3], [4].

Despite its importance, obtaining accurate, continuous measurements of sea-ice thickness at meaningful spatial scales remains a significant unsolved challenge. Satellite-based techniques, employed by missions such as ESA's CryoSat-2 and NASA's ICESat-2, provide broad geographical coverage but rely on indirect inference, estimating ice thickness from the freeboard height above the waterline, and are inherently uncalibrated, requiring in-situ ground-truth data to validate and correct their outputs before they can be considered reliable [5], [6]. Manual methods, like drilling and ice mass-balance buoys, offer high localised precision but are inherently destructive, logistically demanding, hazardous, and expensive to deploy at scale [7]. Autonomous drifting buoys represent a more scalable and cost-effective option for long-duration polar monitoring and have been widely used to record oceanographic parameters in open-water conditions [8]. However, current commercial platforms are not instrumented to measure sea-ice thickness directly, and most are not designed to withstand the prolonged mechanical and thermal stresses of ice-covered polar environments. A thorough review and comparison of existing solutions is provided in Section 1.6.

The lack of reliable data coincides with a period of rapid technological development that now makes autonomous polar monitoring instruments feasible. Advances in low-power embedded systems, compact commercial sensor technology, and electromagnetic measurement techniques present an opportunity to develop a buoy that overcomes the limitations of existing approaches. This project runs alongside research between the University of Manchester and the British Antarctic Survey (BAS) into non-destructive sea-ice sensing methods [9] and seeks to translate that research into a field-deployable prototype.

1.2. Motivation

The scale of the sea-ice thickness monitoring gap becomes clear when examining how rapidly the polar regions are changing. Observational studies have confirmed that the Arctic sea-ice extent has declined at a rate that consistently and significantly exceeds the projections of current climate models [10]. This difference between observation and model output reflects a fundamental deficiency in the quality of the observational data used to calibrate and validate those models. Without accurate sea-ice thickness measurements, the models that scientists depend upon to project future climate change will remain unreliable in precisely the regions that matter most.

There is therefore a clear and pressing scientific need for a monitoring platform that is autonomous, capable of operating continuously over extended deployment periods, and able to measure sea-ice thickness directly and non-destructively. This project was undertaken in direct response to that need, in collaboration with BAS, whose Atmosphere, Ice and Climate team requires exactly this kind of monitoring capability to support their research into atmosphere-ice-ocean interactions [11]. Beyond the immediate scientific application, keeping the unit cost within the budget constraints and maintaining a mass

manufacturable design ensures the platform remains viable for deployment, delivering data that no existing technology can currently provide.

1.3. Aims

The project is characterised by the overarching aim of developing a prototype climate-monitoring sensor buoy suitable for autonomous deployment measuring sea-ice-thickness non-destructively as well as other important environmental parameters in polar regions. The system must be able to demonstrate continuous, reliable data acquisition across a range of polar conditions without requiring human intervention, whilst communicating sensor data over long-term autonomous deployment, employing renewable energy harvesting methods. The prototype should serve as a proof-of-concept platform from which a scalable buoy network could be developed in partnership with BAS.

1.4. Objectives

To achieve the aim, the following objectives are set to ensure a functioning buoy system:

1. Conduct a literature review of existing climate monitoring buoys and sea ice measuring techniques, to inform the system requirements and design decisions.
2. Design and implement an imaging system capable of capturing visual records of the local environment, incorporating image processing prior to transmission as to reduce bandwidth.
3. Design and implement a robust suite of modular sensors featuring compatibility within multiple environments, aligning with existing Arctic climate-monitoring technologies.
4. Develop appropriate Microcontroller Unit (MCU) hardware, capable of acquiring and processing data from respective sensing systems autonomously.
5. Engineer a cold-resilient, energy-harvesting power architecture that balances ultra-low quiescent sleep states with high-current demands for long term deployment.
6. Design and build mechanical packaging within a form factor designed for mass deployment, capable of housing all subsystems in a robust, and manufacturable enclosure.
7. Integrate all developed sensing subsystems into a prototype capable of a field trial.
8. Implement a low data rate communication system for remote sensor data transmission and remote device configuration, representative of satellite communication limitations.
9. Develop a graphical user interface (GUI) ground-station to request, organise, visualise, and analyse environmental data across multiple sensors.
10. Integrate Objectives 8 and 9 with the physical sensing hardware of the buoy to construct a complete, deployment ready climate monitoring system meeting the overall project aim.

1.5. Specification

The following defines the key technical and operational requirements to meet project objectives:

- 1- **Sea Ice Thickness (SIT) measurement** - Develop a non-destructive method of measuring sea ice up to three meters.
- 2- **Accurate ECV monitoring** - Develop a system that can measure a range of novel Essential Climate Variables
- 3- **Power** - Integrate renewable energy harvesting to reliably power sensing and communication for two years deployment, with sufficient battery storage to overcome winter darkness.
- 4- **Air Deployment** - The buoy must be designed such that it can be deployed from air encapsulated in a NATO class B sonobuoy.
- 5- **Temperature rating** - All system components and materials must be rated for operation at temperatures at or below -40°C to ensure suitability for Arctic deployment.

1.6. Review of Existing Technology

Identifying gaps in current technology is essential to achieving Objective 1 and informing the design specifications in this report. This section reviews existing solutions across the two foundational elements of the project: climate-monitoring buoys, and sea-ice thickness measurement.

1.6.1. Climate Monitoring Buoys

Instrumented buoys are an established platform for autonomous climate monitoring, though most are optimised for a single environment. The following examples were selected for mechanical and electrical elements that influenced the design of this project.

The PowerBuoy, is a commercially available autonomous platform combining wave, wind, and solar harvesting to support extended deployment lifetimes in a robust and compact form factor [12]. The hybrid power strategy, modular instrumentation and real-time data transfer demonstrate that long-term autonomous operation at scale is feasible. However, the PowerBuoy provides limited climate-sensing capability which is not optimised to measure polar variables.

Datawell's Waverider buoys provide self-orienting, high-resolution oceanographic sensing, with measurements achieving within 5% of calibration reference values [8]. However, the modules are heavy, weighing up to 150 kg [13] and the platform is restricted to open-water deployment. Downscaling the footprint and adapting equivalent measurement quality in polar environments remains an open venture.

Ice Mass-Balance Buoys (IMB) are the closest existing technology to the project scope featuring sea ice thickness (SIT) measurements and climate sensing capability. SIT is derived from acoustic rangefinders positioned above and below the ice, with additional environmental monitoring modules attached to the main structure. This approach provides optimal remote SIT measurement accounting for both snow and basal ice growth [14], however, the platform is stationary and constrained to thick multiyear ice rendering it inoperable on thinner seasonal ice. Deployment is environmentally disruptive and requires specialist personnel to manually tether the module into the ice [15], increasing operational cost and limiting scalability, evidenced by only 32 deployments since 1993 [14]. This highlights the need for a non-destructive, autonomous alternative device capable of operating across varying ice conditions.

The Seasonal Ice Mass-Balance Buoy (SIMB) addresses the thin-ice limitation present on the IMB enabling operation on seasonal ice [16]. Unlike the IMB, SIMB Instrumentation is non-modular with sensors embedded into the spar, however, its climate-sensing ability is reduced, measuring only temperature and pressure [16]. The platform inherits the same requirement for manned initial deployment and the associated environmental disruption limiting large-area scalability. Despite this, its integrated form factor demonstrates that a self-contained polar sensing with SIT capability is feasible within a tube architecture.

1.6.2. Sea Ice Thickness Measurement

Several existing sea-ice thickness measurement techniques are in use, with differing trade-offs in accuracy, coverage, and operational complexity. The most suitable approach for integration into an autonomous buoy platform is assessed below.

Direct drilling using augers is the calibration reference standard for SIT measurements, providing millimetre resolution accuracy and the ability to distinguish draft and freeboard ice thickness [17]. However, the method is destructive, non-repeatable, time-consuming, and incompatible with thin ice which limits scalability [18]. This method is not viable for the scope of the project as it is environmentally destructive, and operational complications require human intervention.

Upward-looking sonar (ULS) measures sea-ice draft from below sea-level, traditionally deployed on submarines, autonomous underwater vehicles (AUVs) and notably the IMB and SIMB. This provides strong

spatial coverage at a high operational cost [19]. As such, ULS is most viable for wide-area coverage missions and is not suitable for mass deployment due to the high cost and operational complexity.

Satellite technologies such as CryoSat-2 [20] and ICESat-2 [21] are not feasible for direct implementation, however these platforms provide context for this project as the devices would benefit from cross-validation against localised, mass-deployable in-situ measurements, capabilities this project is designed to provide as set out in the aims and specification.

Electromagnetic induction (EMI) exploits the conductivity contrast between seawater and sea-ice to infer thickness from the induced secondary field. This principle is utilised in the airborne HEM-Bird which has an accuracy of ± 0.10 m up to a thickness of 3m against calibration standards [22], with sled-operated ground-based solutions maintains a similar accuracy [23]. However, both systems require active human operation, limiting spatial coverage and scalability. EMI therefore presents the most suitable candidate for adaptation into an autonomous buoy platform, offering non-destructive SIT measurement with a practical balance of accuracy, cost, and operational complexity.

1.7. System Architecture

The system proposed consists of two mechanically and electrically partitioned sections. A rotating assembly holds orientation sensitive components such as the EMI sea-ice thickness measurement coils, cameras and solar panels, in addition to internal systems including battery monitoring and internal temperature sensors, which are all connected to a central microcontroller board. The static side maintains a majority of exteroceptive sensors due to waterproofing and rotational mobility constraints. The block diagram of the system architecture is presented in Figure 1:

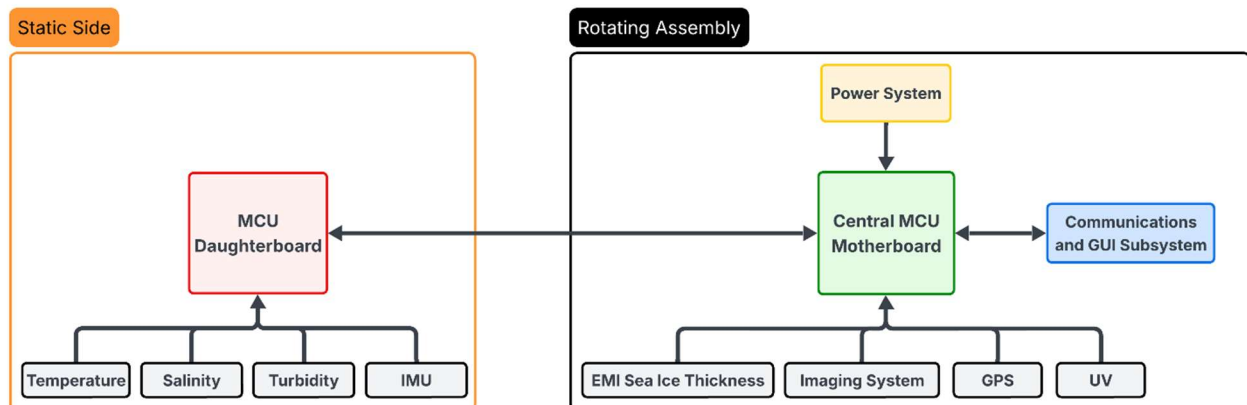


Figure 1 – Proposed system architecture

The completed system is made up of the following elements:

- **Central Microcontroller Motherboard** – The central processing hub of the system on the dynamic partition of the buoy, built around the STM32. Receives, schedules, executes tasks and interconnects all other system features.
- **MCU Daughterboard** – Located on the static side of the buoy and electrically connected to the central motherboard via a slip ring to measure and digitise sensor signals. This aims to reduce signal noise and maintains data integrity of static side sensors.
- **Power Delivery System** – Responsible for battery charging, management and power distribution, integrating a solar charger and appropriate output rails for sensing and communication loads.
- **Communication System** – Low-power Bluetooth-based architecture, mimicking a satellite-based system, responsible for ground station communication and receiving user inputs. Transfers data packets allowing for data capture, sorting and visualisation.

- **EMI Sea-Ice Thickness** – Provides non-destructive sea-ice thickness measurement through an EMI-based sensing architecture that exploits the conductivity contrast between water and sea ice.
- **Imaging System** – Dual cameras which allow for remote image capture, features on-board software for image pre-processing and filtering with metrics to reject low information captures reducing wasted communications bandwidth.
- **Environmental Monitoring** – Additionally integrating remaining environmental sensors in the system across both partitions of the buoy including Temperature, Turbidity, Salinity, IMU wave spectra, UV and GPS.

1.8. Report Summary

The report examines the electronic, mechanical and technical design processes behind the development and construction of the buoy. Each subsystem as outlined in Section 1.7, presents a literature review, key design decisions, followed by implementation and testing procedures. The report then discusses the integration of the buoy's subsystems through full firmware and hardware implementation. Finally, the report evaluates the project outcomes against the original objectives, including reflections on project management and recommendations for future work. Figure 2 presents a labelled cross-sectional view of the prototype buoy, illustrating the integrated system, each of which is referenced and discussed in detail in the sections that follow.

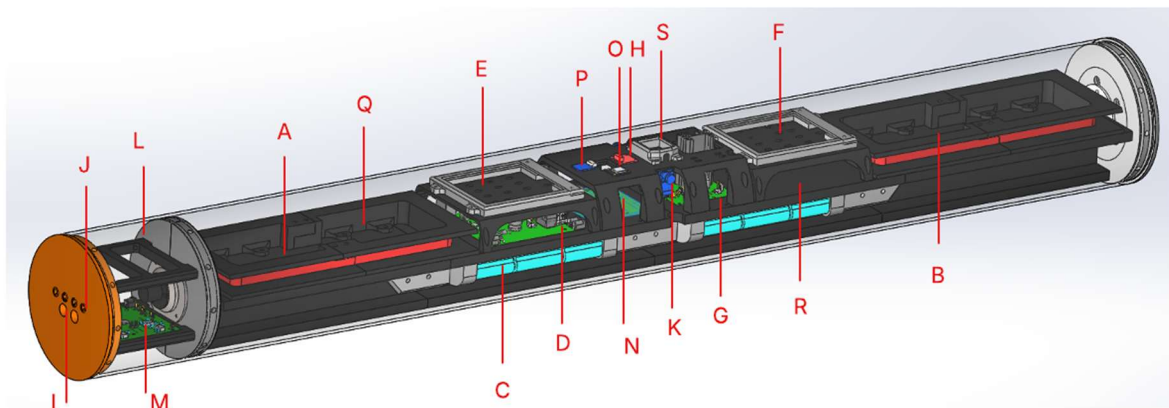


Figure 2 - 3D render of assembled buoy including labelled subsystems

2. EMI

2.1. Introduction

Sea Ice Thickness (SIT) is designated by the World Meteorological Organization (WMO) as an ECV due to its critical role in oceanic heat flux and global climate modelling [24], [25]. While satellite altimetry provides widespread data on ice extent, accurate, high-resolution SIT tracking remains challenging. The buoy addresses this by employing low-frequency Electromagnetic Induction (EMI) to perform non-destructive, autonomous SIT measurements, providing localised ground-truthing data for broader climate models.

2.2. Literature Review

Several non-destructive electromagnetic induction methods have been developed for measuring sea-ice thickness. These methods exploit the strong conductivity contrast between the resistive sea ice and the relatively conductive seawater beneath it. A common approach is frequency-domain electromagnetic induction, such as the Geonics EM31 system [26]. In this method, a sinusoidal current is applied to a transmitter coil, while a receiver coil measures the secondary magnetic field generated by eddy currents in

the seawater. By analysing the amplitude and phase of the received signal, commonly represented by in-phase and quadrature components, the distance to the seawater interface can be inferred and used to estimate total snow-plus-ice thickness.

Frequency-domain systems have demonstrated the use of resonant tuning in the transmitter and/or receiver circuitry. Resonance is not a separate sea-ice measurement principle, but an instrumentation-design method used to improve signal strength and frequency selectivity at the chosen operating frequency [27]. However, resonant systems can be more sensitive to component tolerances, temperature drift and changes in coil geometries. Another important design consideration is direct coupling between the transmitter and receiver coils. Since the primary field from the transmitter is much larger than the secondary field, direct coupling can reduce receiver dynamic range. Bucking or compensation coils can be used to null the primary field before amplification [28]. However, passive bucking also affects the measured secondary response and is required to be accounted for during calibration and modelling [29].

An alternative approach is time-domain electromagnetic induction, where a short current pulse is applied to the transmitter coil and the transient decay of the secondary magnetic field is measured after the pulse has ended [30]. Since the pulse contains a range of frequency components, the decay response contains information about the conductivity and geometry of the surrounding medium. However, time-domain EMI generally requires more complex transient measurement circuitry, and the method is less established than frequency-domain EMI for routine sea-ice thickness measurements.

Based on the comparison in Table 1, this project uses a frequency-domain approach. Due to the size constraints of a NATO sonobuoy, the transmitter (TX) and receiver (RX) coils cannot be separated by several metres as in conventional EM31-style systems. This makes direct TX-RX coupling a significant design challenge. Therefore, the proposed system combines compact coil geometry with resonant receiver tuning to improve sensitivity, while considering compensation and post-processing methods to reduce the effects of direct coupling and drift. The aim is not to fully characterise sea-ice structure, age or composition, but to estimate ice thickness from changes in the received electromagnetic response as the distance to the conductive seawater interface varies.

Table 1 - Comparison of EMI measurement approaches

Approach	Principle	Advantages	Limitations	Relevance
Non-resonant frequency-domain EMI	A continuous sinusoidal TX field induces eddy currents in seawater, and the RX coil measures the secondary field.	Established method for sea-ice thickness. Simpler signal processing than time-domain EMI. Can use in-phase and quadrature components.	Requires suitable coil spacing and calibration. Direct TX-RX coupling can dominate if coils are close together.	Forms the main basis of the proposed system.
Resonant frequency-domain EMI	TX or RX circuitry is tuned to resonate at the operating frequency.	Improves signal strength and frequency selectivity. Can increase sensitivity at the chosen frequency.	More sensitive to drift, component tolerances and changes in coil parameters.	Useful because the compact system needs to maximise RX sensitivity.

Bucking / compensation coil	An additional coil cancels part of the direct primary field at the receiver.	Reduces receiver saturation and improves dynamic range. Helps separate weak secondary fields from large primary coupling.	Adds hardware complexity. Can introduce bias if the bucking coil also senses the seawater response. Needs calibration or modelling.	Relevant because small coil spacing increases direct coupling.
Time-domain EMI	A current pulse is applied and the transient decay of the secondary field is measured.	Provides broadband information from one pulse. Can separate responses by decay time.	Requires fast switching, precise timing and sensitive transient measurement. Less established for sea-ice thickness.	Useful as a comparison, but less suitable for the current compact prototype.

2.3. Coil Geometry Simulation

2.3.1. Simulation Environment

To validate the underlying physics and optimise the geometric constraints of the sonobuoy tube prior to hardware fabrication, the system was modelled using the Low-Frequency Magnetic Quasi-Static (MQS) solver in CST Studio Suite [26]. The environment was configured with a background material of air, a resistive sea ice layer ($\sigma = 0.01 \text{ S/m}$), and a seawater boundary ($\sigma = 2.5 \text{ S/m}$). The system was simulated at an operating frequency of 30 kHz. While the exact final hardware implementation features variations in component tolerances, this environment provides a theoretical baseline for the electromagnetic design.

2.3.2. Coil Orientation Analysis

The primary design decision for the EMI array was the relative spatial orientation of the TX and RX coils. Standard geophysical surveying utilises either a Horizontal Coplanar (HCP) or Vertical Coplanar (VCP) layout.

To determine the optimal configuration for the buoy, the simulation swept the TX and RX coils from an HCP alignment (0°) to a VCP alignment (90°) over a fixed 3-meter ice thickness.

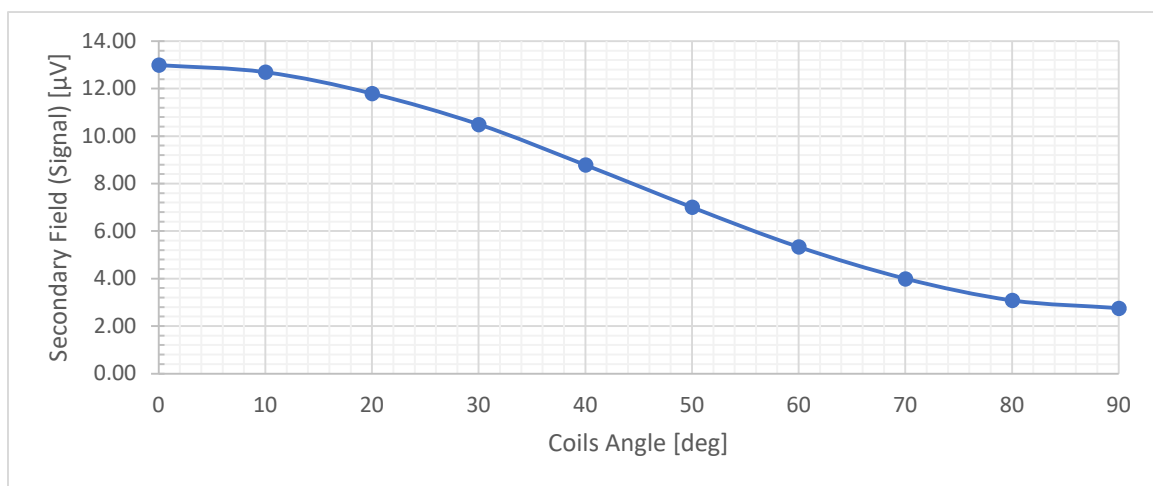


Figure 3 - Effect of Coil Orientation on Secondary Field

As demonstrated in Figure 3, the secondary field magnitude reaches its absolute maximum ($13.0 \mu\text{V}$) at 0° (HCP) and decays significantly to a minimum ($2.76 \mu\text{V}$) as the coils approach 90° (VCP). This aligns perfectly with classical electromagnetic theory, which dictates that at low induction numbers, a HCP dipole

configuration provides approximately twice the effective depth of exploration compared to a VCP configuration due to the optimal angle of incidence of the primary magnetic flux lines with the conductive boundary [31]. Consequently, the mechanical chassis was designed to mount the coils horizontally.

2.3.3. Coil Geometric Optimisation

Commercial EMI systems maximise investigation depth by maintaining large coil areas and significant separation distances (often >3 m). This increased separation forces the magnetic field lines to propagate along more distant paths, ensuring deeper penetration into the environment to couple with the conductive seawater interface; reduced primary-field coupling at the receiver is a secondary byproduct [31].

However, the platform is rigidly constrained by the ~175 mm diameter and internal structure of the sonobuoy tube, requiring both coils to be housed within a 1.3 m length. This introduces a severe engineering trade-off: increasing the coil area by adding length, leads to a rise in magnetic dipole moment but physically forces a reduction in the separation between the TX and RX coils.

To resolve this, a parametric sweep was conducted, expanding the coil length from 0.16 m to 0.60 m.

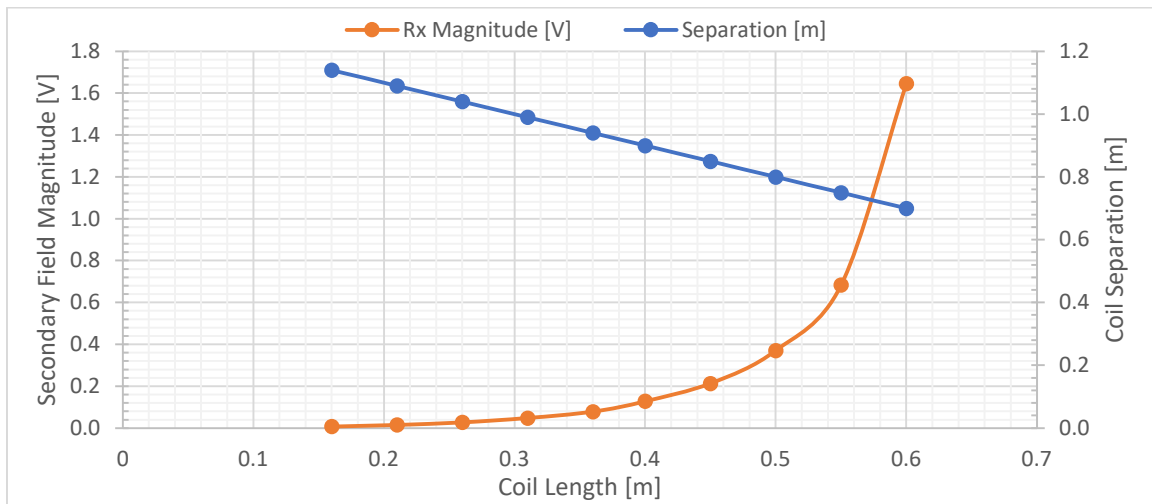


Figure 4 - Trade-off between Coil Area and Separation Distance

As illustrated in Figure 4, the sensitivity gain provided by a larger coil area heavily outweighs the penalty incurred by reduced separation when evaluated at a SIT of 3-metres. Despite the coils moving closer together, the magnitude of the secondary return signal improved by more than an order of magnitude, from 9.00 μV to 0.124 mV, a 1,278% increase. These results confirmed that maximising the coil winding area up to the physical limits of the sonobuoy yields the highest Signal-to-Noise Ratio (SNR), providing the most robust detection of the ice-water boundary within the restricted mechanical envelope.

2.4. Design and Implementation

2.4.1. Coil Design

Following on from the simulations, the TX and RX coils, shown in Figure 2 labelled 'A' and 'B' respectively, were designed to maximise their enclosed area within the constraints of the sonobuoys 174.7 mm diameter. A parametric model was made in SolidWorks to allow the coil geometry to be changed whilst the model managed the dynamics of the tube constraint, primarily the changing in maximum width as coil height was increased. Figure 5 shows the model's initial drawing and the subsequent assembly of the two coil halves. The coils were 3D printed using PLA filament, bonded together using a two-part epoxy and then wound as the next report section will discuss.

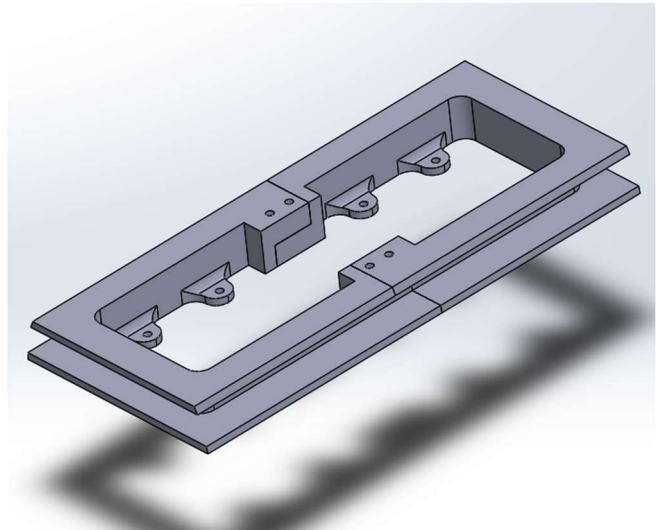
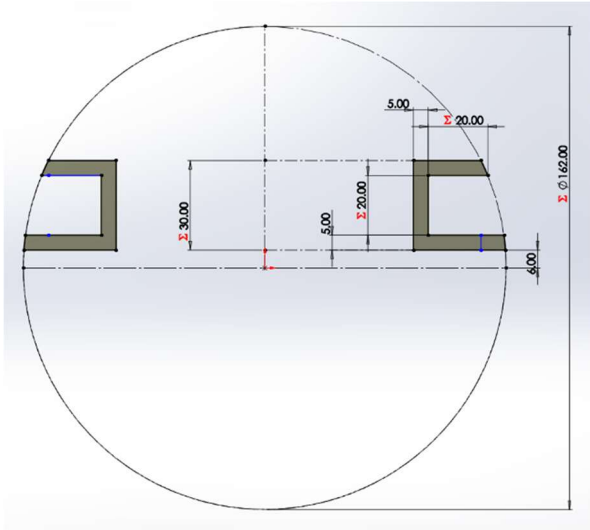


Figure 5 - Parametric model and coil design

Resonant Tuning

The minimum number of wound turns for the TX coil was estimated from the drive capability of the OPA541 power amplifier. From the datasheet, the minimum recommended load impedance was approximately $5\ \Omega$. Therefore, Wheeler's formula for rectangular coils was used to estimate the number of turns required to achieve a suitable coil impedance. This gave an estimated value of 10 turns. To allow for calculation uncertainty and manufacturing variation, the TX coil was designed with 13 turns. A thicker wire gauge was also used for the TX coil compared with the RX coil, as the TX coil is required to carry a larger drive current.

Shifting to the RX coil, impedance spectroscopy was carried out to characterise its resonant behaviour and develop an equivalent circuit model. The measured impedance magnitude response is shown in Figure 6. The response shows a clear resonant peak, indicating that the RX coil behaves as a parallel resonant system due to the interaction between the coil inductance and parasitic winding capacitance.

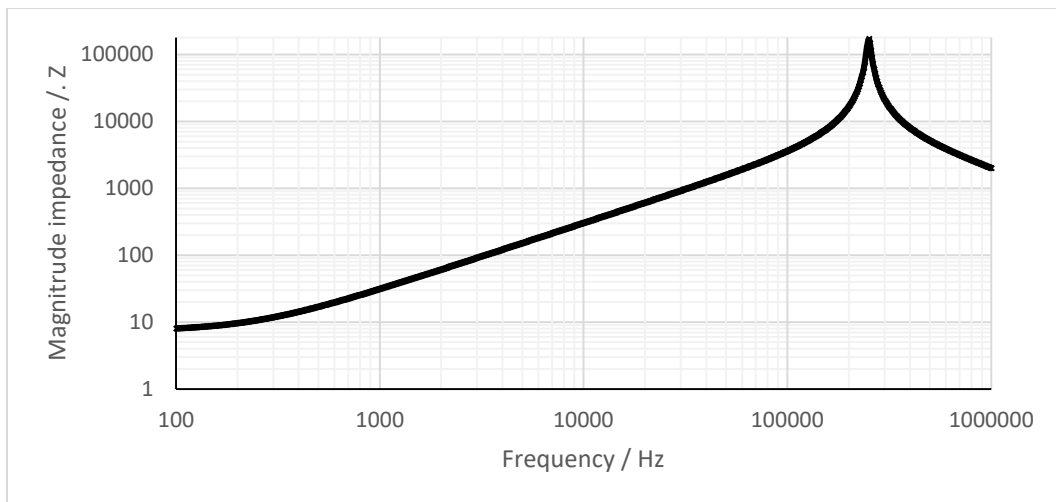


Figure 6 - Impedance Magnitude Response of Resonant Coil

The impedance spectroscopy data was exported to MATLAB and used to estimate the equivalent circuit parameters. The model consisted of a series resistance (R_s), an inductance (L), a parallel capacitance (C_p), and a parallel resistance (R_p). The equivalent impedance of the model is given by:

$$Z(\omega) = \frac{1}{\frac{1}{R_p} + j\omega C_p + \frac{1}{R_s + j\omega L}} \quad (\text{Equation 2.1})$$

The series resistance was estimated from the DC impedance value of the measured response, where the coil is dominated by winding resistance. This gave a value of approximately 8 Ω . The parallel resistance was estimated from the impedance magnitude at the resonant peak, where the inductive and capacitive reactance cancel, and the impedance is dominated by the parallel resistance. This gave a value of 171.16 k Ω . The equivalent circuit parameter was found in MATLAB using system identification method.

Table 2 - RX coil equivalent circuit parameters

Parameter	Value
Series Resistance	8 Ω
Parallel Resistance	171.16 k Ω
Parallel Capacitance	83.22 pF
Inductance	4.82 mH

The model was used to determine the capacitance required to shift the RX coil resonance down to the desired operating frequency of 19.8 kHz. To shift resonance to 19.8 kHz, an additional parallel capacitance was added to the equivalent circuit. The required capacitance was calculated using Equation 2.2.

$$C_1 = \frac{1}{4\pi^2 f^2 L} - C_p \quad (\text{Equation 2.2})$$

Using the fitted inductance value of 4.82 mH and the target frequency of 19.8 kHz, the required additional capacitance was calculated as approximately 13.3 nF. A parallel damping resistor was also added to reduce the quality factor of the resonance. This broadens the resonance peak, increasing the usable bandwidth and reducing sensitivity to small variations in component values, coil parasitics, and environmental loading. The damping resistance was calculated using the quality factor relationship for a parallel resonant circuit:

$$Q = \frac{R_p}{\omega_0 L} \therefore R_{p,new} = Q\omega_0 L \quad (\text{Equation 2.3})$$

A parallel resistance of 12 k Ω was selected to provide a broader and more stable resonance around the 19.8 kHz operating frequency. The modified equivalent circuit was then simulated in LTspice using the added 13.3 nF capacitor and 12 k Ω damping resistor. The resulting shifted impedance response is shown against the normal impedance response in Fig. 8, and the corresponding LTspice equivalent circuit is shown in Fig. 7.

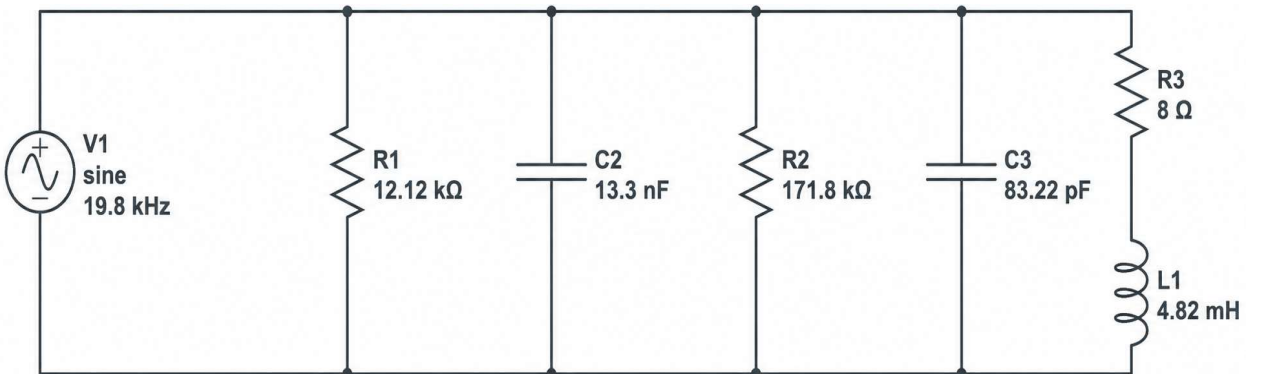


Figure 7 - Equivalent circuit of coil including extra parallel resistance and capacitance values

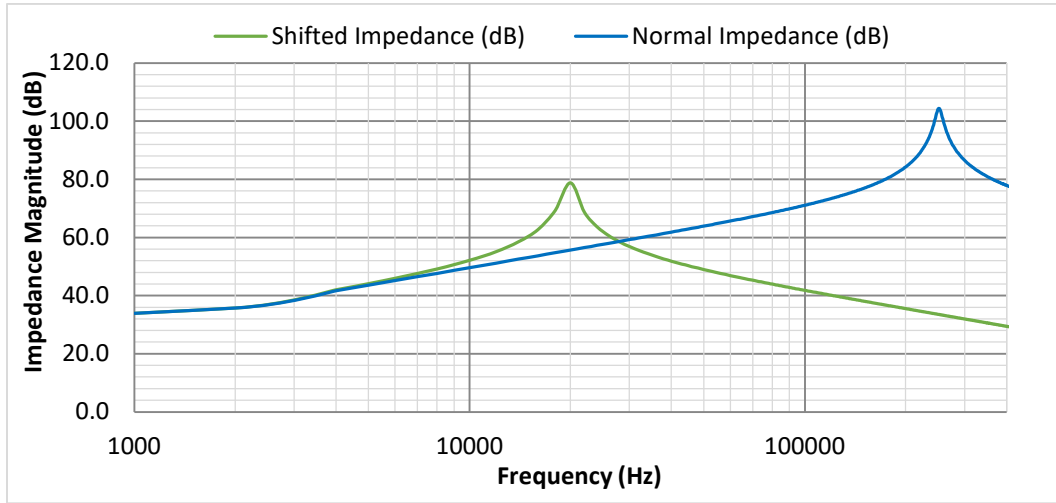


Figure 8 – Pre and post resonance tuning of RX coil

Bucking Coil Implementation

The direct inductive coupling between the TX and RX coils produces a primary field voltage at the RX coil that significantly dominates the secondary field contribution from the seawater target. Without mitigation, this coupling would drive the RX coil voltage to approximately 1.5 V, masking the target signal entirely and risking saturation of the downstream amplifier stages. A bucking coil was therefore wound and placed inside the TX coil, as shown in Figure 2 labelled 'Q', wound in the opposition direction to the RX coil to provide physical cancellation of the primary field before amplification.

Experimentally, the bucking coil was found to provide optimal cancellation when wound as a single loop 5 cm in diameter, with twisted pair wiring to the RX coil. Due to the sensitivity of the cancellation performance to changes in geometry, it was wound tightly around a 3D printed former and securely mounted to the TX coil such that it could not shift within the field and alter the coupling.

2.4.2. Electronic Design

The EMI electronics are critical to the sensitivity and resolution of the measurement system. To allow independent development and testing before integration with the main buoy electronics, the transmit and receive circuitry were implemented on a dedicated two-layer PCB. The design used analogue layout practices, including ground planes, shielding, and stitching vias, to minimise return path lengths, improve signal integrity, and reduce susceptibility to interference.

Transmit Coil Drive Circuit

To drive the transmit side, an OPA541 power amplifier was chosen; with a peak current rating of 10 A and a bandwidth of 1.6 MHz, it provides sufficient headroom for operation at 20 kHz [32]. The amplifier was driven by the STM32's digital-to-analogue converter (DAC), responsible for generating a 1.1 V peak-to-peak at 19.8 kHz sine wave signal, where it is powered from the +15 V and –15 V rails of the power system.

As shown in Figure 9, the signal input from the DAC was AC-coupled through a 100 nF ceramic capacitor to remove the DC offset from the DAC output. A gain of 22.7 was applied to the signal; through experimentation, this was found to produce the best balance of output voltage without introducing distortion to its waveform. The amplifier was configured with a current limit of 1.5 A, preventing power system collapse in the event of a short circuit during deployment. The power amplifier was decoupled as per the datasheet recommendation using both 100 nF ceramic and 10 μ F electrolytic capacitors [32].

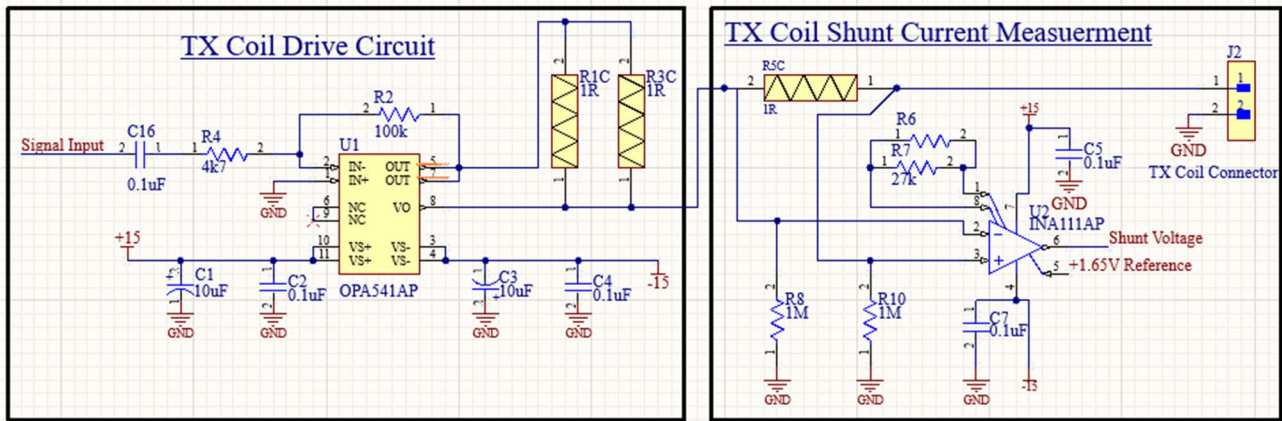


Figure 9 – EMI Transmit Power Amplifier

The drive circuit also includes a 1 Ω series shunt resistor to measure the transmit coil current, as shown in Figure 9. The shunt voltage was measured using an INA111AP instrumentation amplifier with a gain of 3 and a 1.65 V reference, allowing the bidirectional AC current waveform to be sampled by the STM32 ADC. The 1.65 V low-impedance reference was generated using an OPA277PA precision op-amp, derived from the STM32 ADC reference through a precision resistor divider, as shown in Figure 10.

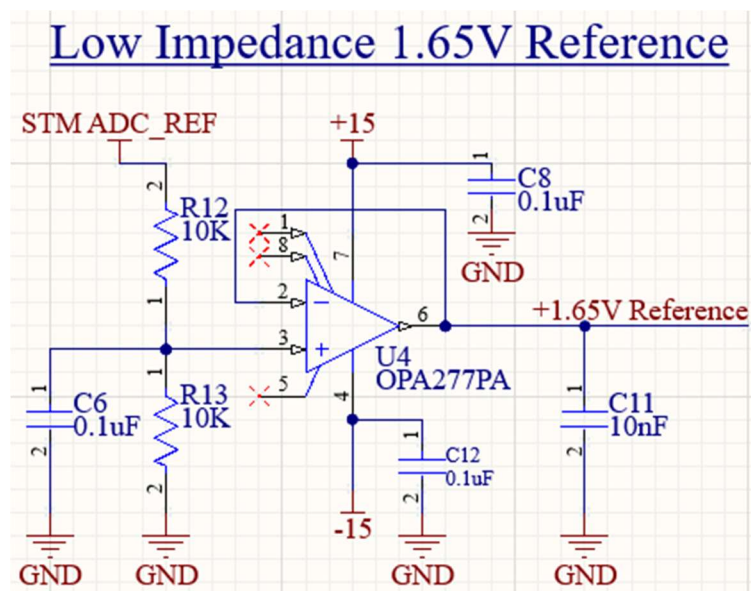


Figure 10 - 1.65 V low impedance reference schematic

Receive Coil Signal Conditioning Circuit

The receive coil signal conditioning circuit amplifies the RX signal while ensuring the ADC input range is not exceeded. A bucking coil was used to reduce the primary coupling from the TX coil, making the RX coil more sensitive to the secondary field generated by eddy currents in the seawater. The RX signal was then passed through an INA111AP instrumentation amplifier, providing improved common-mode rejection and a gain of 17.67 to increase the signal-to-noise ratio. The output was level-shifted for single-supply ADC sampling and kept within the safe measurable range of the ADC. The INA111AP was selected for its 2 MHz bandwidth, allowing the required gain to be achieved in a single stage [33]. The full RX coil circuit is shown in Figure 11.

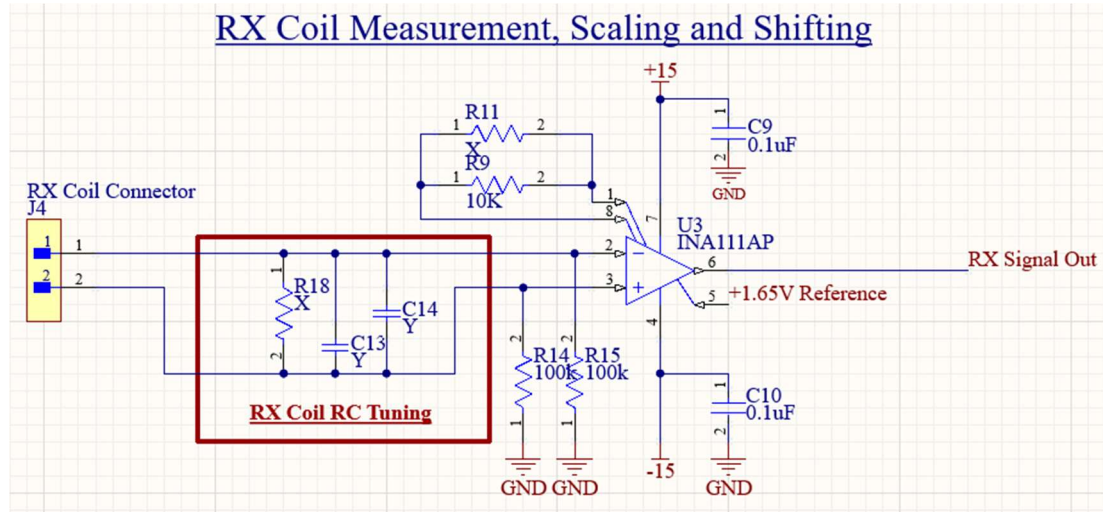


Figure 11 - RX coil circuit schematic

As shown in Figure 11, the resonance tuning components were placed in parallel between the RX coil connector and the instrumentation amplifier. Additionally, 100 k Ω resistors were connected to ground on the amplifier inputs to provide a current path for the bias currents. Due to resonance of the RX coil and significant direct coupling between the TX and RX coils, a gain of 3 was determined to be optimal for maximising the ADC input range. As with the TX shunt measurement circuit, a low-impedance 1.65 V reference was applied to offset the output to the ADC midpoint.

2.5. Data Acquisition and Signal Processing

The EMI board, labelled 'G' in Figure 2, is controlled by the STM32 Motherboard, labelled 'N'. The STM32 generates a 19.8 kHz DAC waveform to drive the TX power amplifier and simultaneously samples the amplified RX coil voltage.

A timer was used to update the DAC from a 128-point sine lookup table and trigger the ADC sampling. This ensured a fixed phase relationship between the transmitted waveform and the measured RX response. The 128-point lookup table provided a balance between waveform resolution and DAC update rate, with a maximum step size of 8.66 mV.

After digitisation, the RX signal is processed to extract the in-phase and quadrature components relative to the TX excitation, as shown in Figure 12. The first stage rejects the initial transient response. During testing, the system settled within approximately 5 ms, equivalent to the first 100 cycles. Therefore, the first 12,800 samples of the 38,400-sample capture were discarded before processing

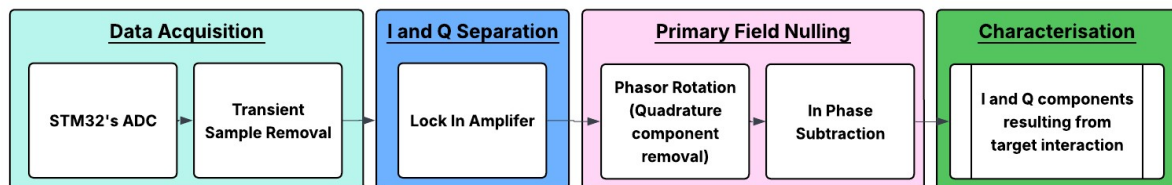


Figure 12 – Data acquisition and processing pipeline

A discrete lock-in amplifier was then applied to the remaining samples to isolate the 19.8 kHz RX component and calculate the in-phase and quadrature values. The reference phase was tuned experimentally to align with the measured TX drive waveform rather than the leading DAC waveform. A phasor rotation was then applied to minimise the quadrature component of the primary field, followed by in-phase subtraction to remove the remaining direct TX-to-RX coupling.

This primary field cancellation is necessary because the direct coupling from the TX coil is significantly larger than the secondary field produced by eddy currents in the seawater target. Removing this dominant component allows small changes to be directly attributed to the target's presence.

2.6. Testing

2.6.1. Methodology

The EMI system was tested in a controlled indoor environment using a 2.50 m × 1.00 m conductive target. The target consisted of a 100 mm thick polystyrene foam board covered with two layers of aluminium foil to create a large conductive surface. To reduce the influence of nearby structural metal, the I-beam assembly was suspended from the ceiling and held vertically, keeping it away from the steel floor structure while allowing the target distance to be adjusted easily. The test setup is shown in Figure 13.

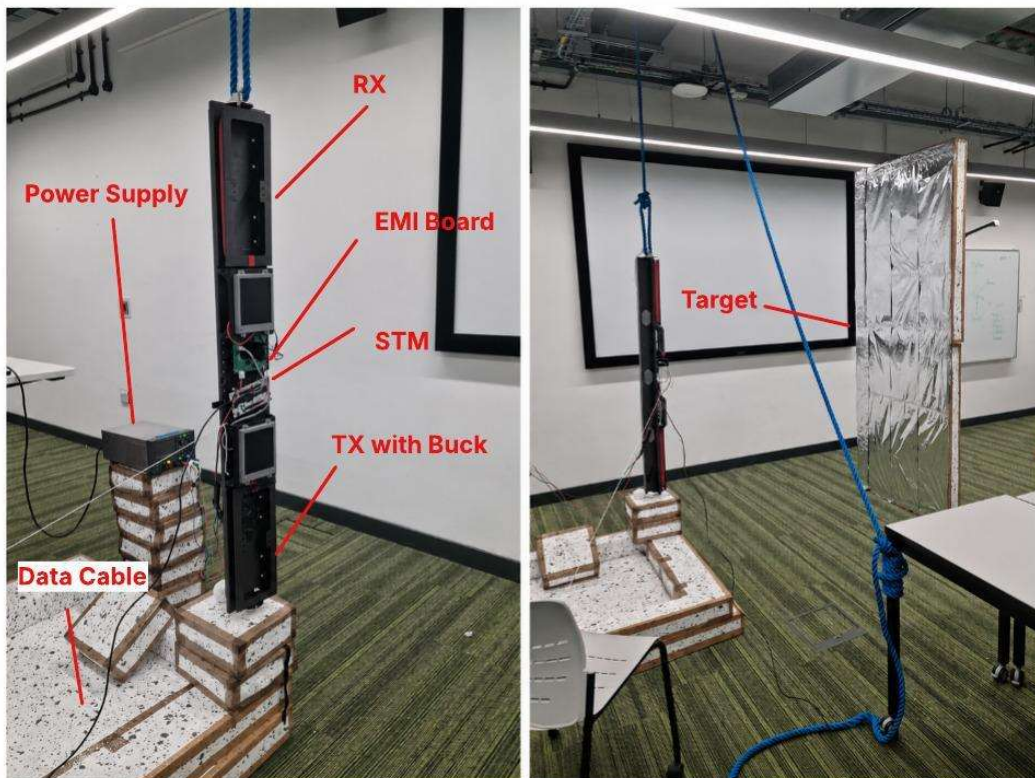


Figure 13 - EMI experimental setup

During testing, the target was moved to set distances from the midpoint of the coil arrangement. At each distance, the firmware and signal processing method described in Section 2.5 were used to extract the in-phase and quadrature voltage components, along with the signal magnitude and phase.

Each distance was tested three times and the results were averaged. Since the firmware also averages multiple captures, each reported value represents the mean response of 150 measurements, with each measurement taken over 200 driven cycles. This provided sufficient averaging to reduce random noise and demonstrate the measurement trends of the system.

2.6.2. Results

Figure 14 shows the averaged in-phase and quadrature components of the amplified RX coil signal after primary field cancellation. At close range, the conductive target produces a strong response, with the in-phase component reaching -2.27 V at 100 mm before changing sign and peaking at approximately 0.74 V at 220 mm. This sign reversal is caused by the coil geometry and changing phase relationship of the induced

eddy current response as the target distance increases. The quadrature component peaks at approximately 1.9 V at 155 mm before decaying with distance. Beyond approximately 400 mm, both I and Q components reduce toward zero, showing the weakening target contribution.

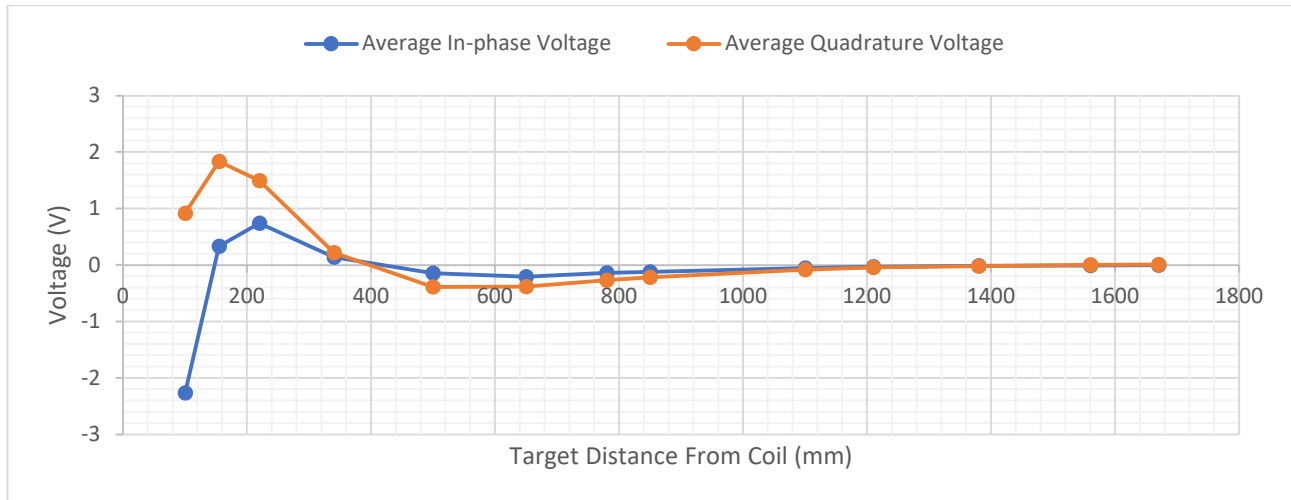


Figure 14 – Amplified RX coil in-phase and quadrature components

Figure 15 presents the corresponding signal magnitude and phase. The magnitude shows the clearest trend, decreasing from 2.44 V at 100 mm to approximately 8 mV at 1670 mm. Most of this reduction occurs within the first 340 mm, after which the response gradually flattens as the target moves further into the far field. This monotonic decay makes magnitude the most suitable output for estimating target distance, as it provides a single-valued relationship with standoff distance.

The phase changes rapidly in the near field, falling from approximately $+158^\circ$ at 100 mm to around -110° by 500 mm. Between 500 mm and 1380 mm, the phase remains relatively stable, suggesting the system is operating in a consistent electromagnetic regime over this range. At the largest standoff distances, notably around 1670 mm, the phase becomes unreliable as the signal magnitude approaches the noise floor.

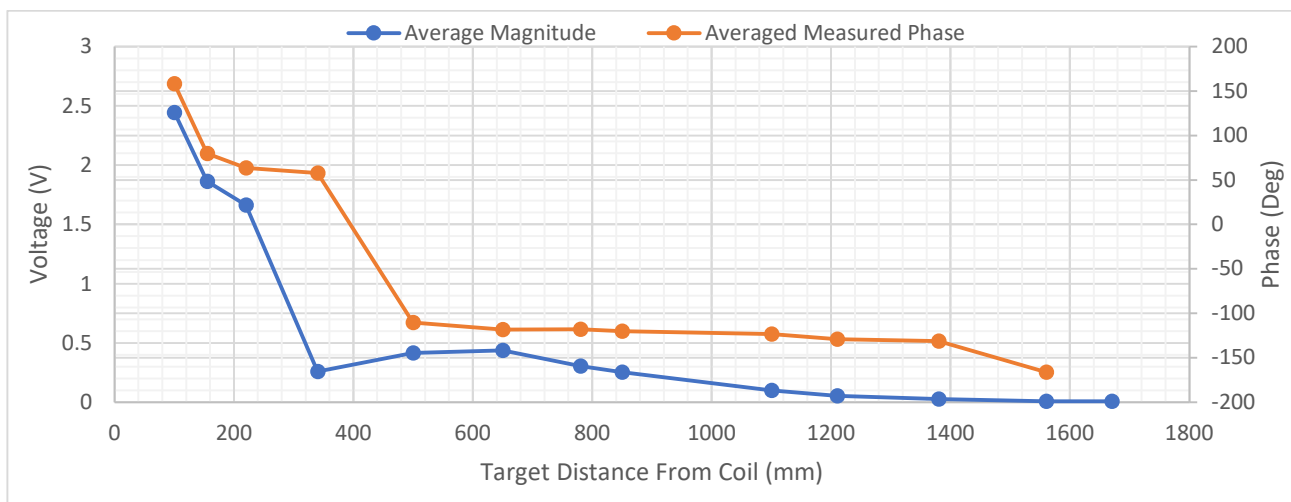


Figure 15 – Amplifier RX coil voltage magnitude and phase relative to the TX coil

Overall, the results show that the EMI system can detect the aluminium target from 100 mm to approximately 1380 mm. Beyond this range, the small signal magnitude reduces confidence in the phase measurement. The magnitude response provides the most reliable channel for future calibration, while the I and Q components remain useful for understanding the electromagnetic response and validating the primary field cancellation method.

2.7. Conclusions

The EMI system has demonstrated the ability to induce eddy currents into a target and measure a resulting voltage on a resonating RX coil. Additionally, with the use of a bucking coil in series with the RX coil and a lock-in amplifier, the system demonstrates the ability to null primary field components and resolve the secondary field components at 19.8 kHz. Given the success of the signal conditioning, further development is limited by the absence of a real seawater target; therefore, future work will explore obtaining higher quality testing data and refining the bucking coil to allow a greater RX gain to be applied.

3. Power and Solar

3.1. Introduction

The power system must sustain autonomous operation in extreme polar environments, enduring -40°C temperatures and 105 days of continuous winter darkness [34]. Mechanically constrained to a strict 33 cm² cross-sectional area along the NATO sonobuoy tube, the hardware must not interfere with the internal self-levelling I-beam chassis (discussed in Section 10). Electrically, it must bridge a vast dynamic load range, maintain an ultra-low quiescent sleep state, while reliably delivering high-current transient bursts for the Electromagnetic Induction (EMI) coils. This chapter details the cold-resilient, solar-harvesting power architecture designed to resolve these constraints and ensure multi-season autonomy [35], [36].

3.2. Literature Review

Traditional polar platforms (e.g., UpTempO buoys [37] and Ice Mass Balance instruments [38]) rely on primary Lithium-Thionyl Chloride (Li-SOCl₂) batteries for their cold-weather resilience. While offering excellent low-temperature discharge, these restrict the platform to a single lifecycle, ultimately becoming hazardous environmental waste. Achieving multi-season autonomy therefore requires a rechargeable architecture paired with energy harvesting.

Table 3 - Comparison of battery chemistries' cold performance and energy densities

Chemistry	Cell Type	Volumetric Energy Density	Low-Temp Discharge (-50°C)	Multi-Season (Rechargeable)	Sub-Zero Charging Risk
Li-SOCl ₂	Primary	~1200 Wh/L	Excellent	No	N/A
Lead-Acid (SLA)	Secondary	~100 Wh/L	Poor	Yes	Low (Electrolyte Freezing)
Lithium Titanate (LTO)	Secondary	~250 Wh/L	Excellent	Yes	None
LiFePO ₄	Secondary	~350 Wh/L	Poor	Yes	Moderate
Li-Ion (NCA/NMC)	Secondary	~700 Wh/L	Moderate	Yes	High (Lithium Plating)

Sealed Lead-Acid (SLA) cells are prohibitively heavy, and Lithium Titanate (LTO) lacks the volumetric energy density (~250 Wh/L) to meet winter capacity limits, therefore Lithium-Ion (NCA/NMC) chemistry was selected for its superior density (~700 Wh/L), allowing sufficient capacity to fit within the 33 cm² I-beam constraint [39].

However, charging Li-Ion cells at sub-zero temperatures prevents lithium ions from successfully intercalating into the graphite anode. Instead, ions accumulate as metallic deposits, causing rapid, irreversible capacity loss and creating severe internal short-circuit risks via dendrite formation [40], [41]. Consequently, the buoy mandates a strict hardware-enforced "cold-charge policy," physically inhibiting the solar charging pathway below 0°C. This dictates the battery pack must be accurately sized to survive the entire 105-day winter purely on stored capacity [34].

A 2S (nominal 7.4 V) architecture was selected as the optimal median voltage. This minimises the step-up/step-down voltage difference for the 3.3 V logic and ± 15 V EMI rails, while maintaining a safe margin below the solar array's maximum power point. Minimising this input-to-output conversion ratio limits extreme duty cycles and reduces switching losses, optimising overall DC-DC conversion efficiency [42].

To replenish this storage architecture, environmental harvesting is required. While wave energy converters offer continuous generation, they rely on moving parts vulnerable to harsh polar conditions [43].

Consequently, photovoltaic (PV) systems are preferred, consistently delivering the highest average power density for remote wireless nodes [44]. Although solar irradiance is lower in polar regions, cold temperatures inherently improve monocrystalline panel efficiency ($\sim 0.52\%/^{\circ}\text{C}$) and increase open-circuit voltage (V_{oc}) ($\sim 0.48\%/^{\circ}\text{C}$) below the standard test conditions (STC) of 25°C [45], [46], [47]. This thermal advantage partially compensates for the weak polar sunlight, ensuring PV remains a viable solution.

Because PV panels exhibit a nonlinear current-voltage relationship, the voltage producing maximum power fluctuates significantly with the buoy's shifting temperature and illumination [48]. Fixed-voltage chargers force sub-optimal operation under these dynamic conditions; therefore, fractional V_{oc} Maximum Power Point Tracking (MPPT) is utilised. This method has been proven in literature to achieve up to 95% tracking efficiency with minimal computational overhead, actively maintaining the panel near its MPP [49].

3.3. Design and Implementation

3.3.1. Power Budgeting and Sizing

The system's endurance is governed by a 105-day winter energy envelope where environmental harvesting with solar is impossible. A comprehensive worst-case power budget, shown in Figure 16, dictates an absolute minimum requirement of 91 Wh. Accounting for severe Li-Ion cold-derating at -40°C , which halves usable capacity [41], this mathematical model dictated a full-scope 2-series, 12-parallel (2S12P) battery configuration using high-capacity 21700 cylindrical cells to survive the winter while fitting the mechanical footprint. (Note: Prototype validation utilises a scaled 2S2P 18650 configuration alongside a 3D-printed mock-up to physically verify the true 2S12P pack dimensions within the full mechanical assembly).

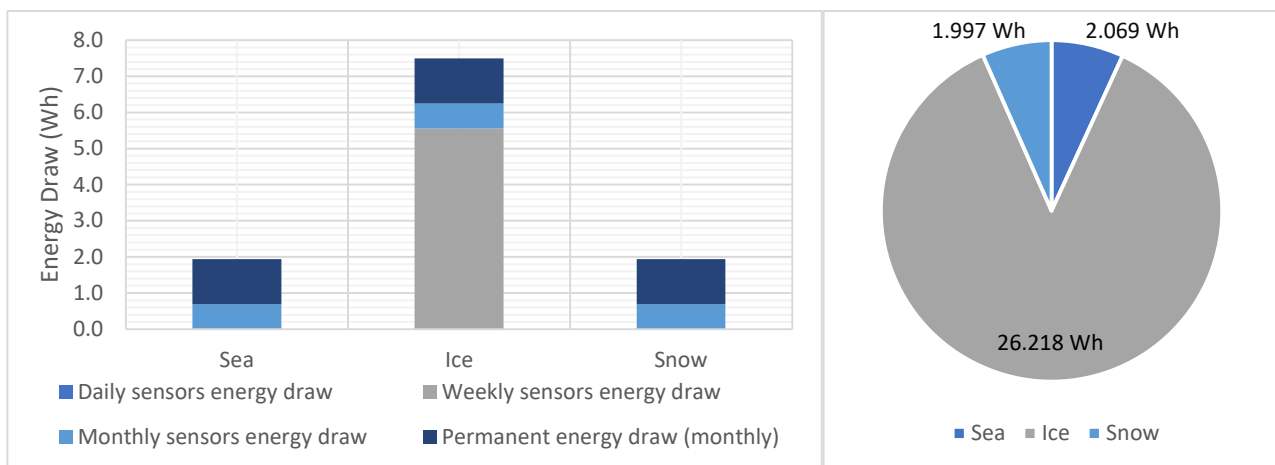


Figure 16 – Comparison of Energy Draw per Environment - draw per duty cycle (left) & total monthly draw (right)

3.3.2. Power Architecture and Output Rail

The distribution network employs a 2S (nominal 7.4 V) architecture routed through a physical master switch, with the final full-scope design incorporating a battery management system (BMS) for active cell balancing [50]. To efficiently support the system's dynamic loads while optimising overall DC-DC conversion efficiency [42], the architecture synthesises four distinct voltage rails:

Table 4 - Power system output rails

Power Rail	Description
<i>Always-on 3.3 V (Buck):</i>	High-efficiency converter to supply the continuous low-current demands of the MCU, RTC, and battery monitor.
<i>Switched 3.3 V (Buck):</i>	Power-gates local sensing systems via enable (EN) pin.
<i>Switched 5 V (Buck):</i>	Dedicated to the Iridium satellite transceiver. (Note: This rail is omitted in the prototype, as telemetry is emulated via a low-power Bluetooth module).
<i>Switched ± 15 V (Boost & Isolated DC-DC):</i>	Drives the active EMI rails. The +15 V boost converter enable pin is physically pulled down to ground to prevent accidental firing during a system reset. In the full-scope design, this rail includes an eFuse for overcurrent protection and gating. The symmetrical -15 V rail is synthesised by cascading a 15W galvanically isolated DC-DC converter directly from the +15V output, allowing 1A per rail.

Because the master switch is mechanically located on the dynamic half of the buoy, a secondary "soft power switch" is implemented on the more accessible static side as a reachable solution through the slip ring. The enable pin for the always-on logic converter is pulled up to the battery voltage, allowing the entire system to be safely disabled by shorting this pin to ground.

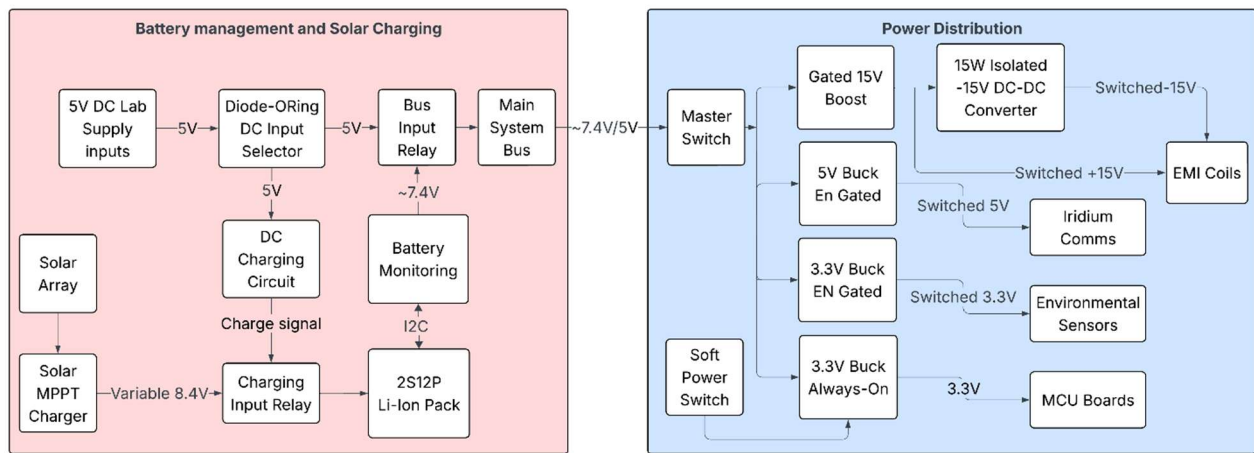


Figure 17 - Power management, distribution and charging architecture

3.3.3. Multiplexing and Hardware Fail-Safes

Adapted from [51], [52], a dual-relay multiplexing circuit safely switches the system between autonomous power (battery/solar) and an external 5V laboratory supply. To ensure a reliable fail-safe, the battery and solar lines are wired directly to the normally closed (NC) relay contacts, forcing the system to mechanically default to its autonomous state if a relay coil fails in the field. Finally, a 2200 μ F bulk capacitor follows the master switch to sustain the voltage rail and prevent MCU brownouts during the physical relay switchover.

3.3.4. Battery Management System (BMS) and SoC Telemetry

The full-scope 2S12P pack necessitates a dedicated hardware TI BQ76905 battery management system (BMS) for active cell balancing and fault protection [50]. For prototype validation, management is scaled to an INA219 current/voltage monitor paired with a custom C++ complementary filter to provide accurate state-of-charge (SoC) telemetry. To eliminate standard Coulomb-counting sensor drift, the algorithm integrates current over time during active loads but anchors the integrated consumption to an estimated voltage-curve SoC during quiescent sleep (<50 mA), applying a 1% corrective pull. Furthermore, the software dynamically derates capacity by interpolating SoC voltage points between a 25°C baseline and a -20°C minimal-operating curve as ambient temperatures drop.

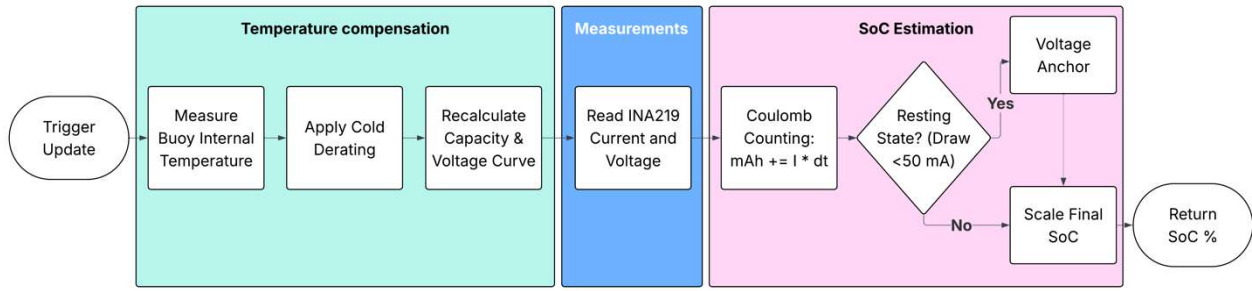


Figure 18 - SoC estimation algorithm including drift correction and dynamic cold-derating.

3.3.5. Solar Array Sizing and Selection

Constrained by the sonobuoy surface area, two Voltaic P126 (6 V, 2 W) ETFE monocrystalline panels are connected in series; mounting on the buoy is represented in Figure 2 under designators E and F [53]. Its IEC 61215 and MIL-STD-810H environmental qualifications ensure resilience during airdrop deployment and harsh polar exposure [53]. Rated at 21.5% efficiency, each provides 2.4 W (STC) with an MPP voltage of 6.84 V and MPP current of 0.3 A [54]. A critical consideration is the cold temperature V_{OC} rise, which defines the panels maximum output voltage and informs the new MPPT setpoint for peak performance in polar conditions. This can be calculated using the cell's temperature coefficient ($\beta_{V_{OC}} \approx -0.239\%/^{\circ}\text{C}$) [55]:

$$V_{OC}(T) = V_{OC,STC} \times [1 + \beta_{V_{OC}} \times (T - 25)] \quad (\text{Equation 3.1})$$

At 5 °C, the 2S string V_{OC} rises from 16.68 V to 17.68 V, safely below the charge controller's 32 V ceiling, while MPP voltage rises from 13.68 V to 14.33 V [56].

During the polar summer, the buoy receives weak, low-angle sunlight at typical irradiance levels of 100-400 W/m², with around 4 peak sun hours each day [57]. Under these conditions, each 2S string generates approximately 3.35 W, yielding a total of 13.4 Wh/day to replenish the system before returning to complete reliance on the battery supply during the polar winter.

3.3.6. Charge Controller

Table 5 below compares the considered charge controllers:

Table 5 - Comparison of Charge Controllers

Features	LT3652 (Analog Devices)	BQ24650 (Texas Instruments)	MCP73842 (Microchip)
Topology	Switching Buck	Synchronous Buck	Linear
Solar / MPPT Suitability	Fixed voltage MPPT via VIN_REG pin	Fixed voltage MPPT via MPPSET pin	Not suitable, no MPPT
Input Voltage Range	4.95 – 32 V	5 – 28 V	8.7 – 12 V
Max Charge Current	2 A	10 A	1 A
Operating Temperature	-40 °C to +125 °C	-40 °C to +125 °C	-40 °C to +85 °C
Battery Chemistry	Li-Ion/LiPo/SLA	Li-Ion/LiPo, Lead Acid	Li-Ion/LiPo only
External Components	Inductor, capacitors, resistors, diodes, thermistor (~ 12)	MOSFETs, inductor, capacitors, resistors, diodes, thermistor (~ 22)	MOSFET, capacitors, resistors, thermistor (~ 8)
Typical Efficiency	~ 75 – 85 %	~ 80 – 95 %	~ 50 – 70 %
Cost	£9.60	£3.49	£1.59

While the MCP73842 is the simplest of the three to implement, it is excluded as the 2S panel string voltage exceeds its 12V maximum input [58]. It also offers no MPPT capability, which is crucial for extending the

buoy's lifespan. Both the BQ24650 and LT3652 satisfy this requirement [56][59]. However, the LT3652 IC was ultimately selected due to the additional complexity of building the circuit for the BQ24650. The LT3652's MPPT setpoint is programmed through an external resistor divider at the VIN_REG pin, with the target voltage derived from the V_{mp} calculations in Section 3.3.5. The maximum charge current, battery float voltage, and temperature protection thresholds are also set through external resistor networks and an NTC thermistor. The values were chosen based on datasheet recommendations and the specific operating conditions of the buoy [56].

3.3.7. Power PCB Implementation

To physically validate the distribution and charging architecture, a PCB was fabricated shown in Figure 2, under designator D. The prototype integrates breakout modules (e.g., Pololu regulators, Adafruit INA219) soldered directly to the board. This implementation successfully unites the LT3652 MPPT network, dual-relay multiplexing fail-safes, the always-on 3.3 V logic supply, and the gated ± 15 V EMI converters, proving the viability of the isolated rail generation and soft-power switching. While currently scaled to a 2S2P configuration, the architecture is scalable. For final field deployment, the system will expand to the full 2S12P capacity alongside the advanced hardware BMS, eFuses, and iridium power rail.

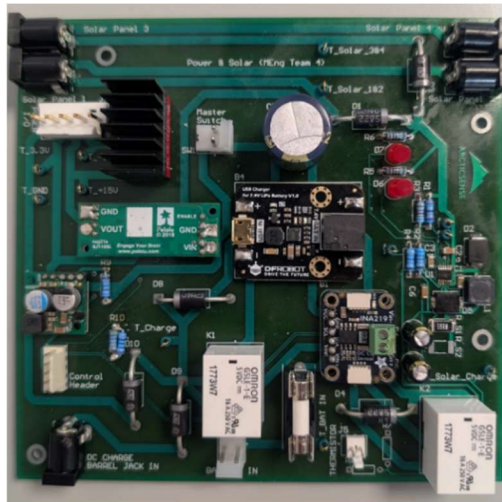


Figure 19 - Physical implementation of the power distribution and solar MPPT PCB

3.4. Testing

A critical sensor network failure mode is a "brownout" reset, where high-power payload activation (e.g., the ± 15 V EMI rails) sags battery voltage, crashing the primary 3.3 V line. To assess this risk, a 4-channel oscilloscope captured a transient load analysis simulating a brownout event, measuring the 2S battery input, 3.3 V logic rail, ± 15 V outputs, and EN trigger response.

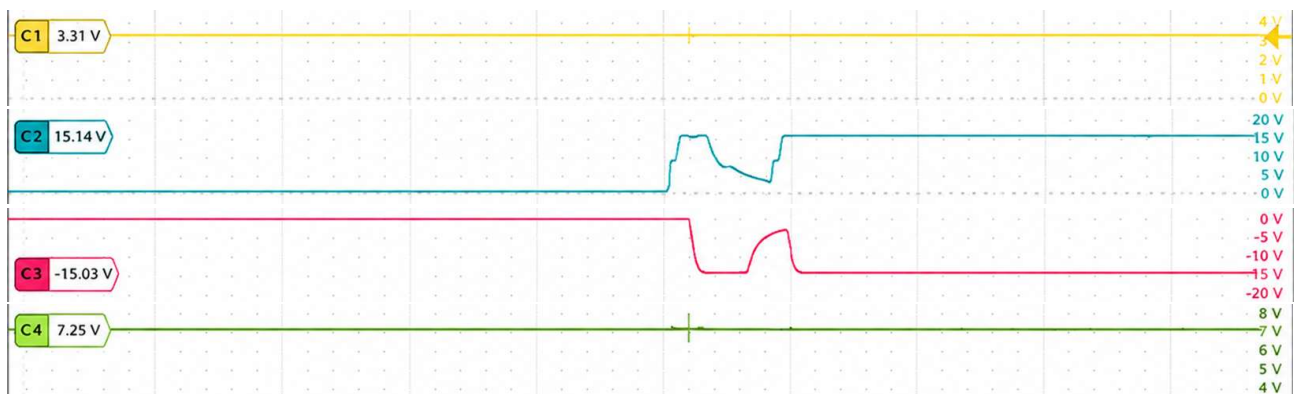


Figure 20 - Oscilloscope data showing always-on 3.3V (C1), ± 15 V outputs (C2 & C3) and 2S battery input (C4)

Upon load connection (~200 ms), the EN pin is driven high, and the 15 V boost converter activates cleanly, reaching steady-state within 50 ms. Crucially, during this transient spike, the 3.3 V logic rail and 2S battery input exhibit a negligible voltage droop, maintaining stable regulation. This physical validation confirms the EMI payload is electrically isolated from critical circuitry, guaranteeing autonomous sensing cycles without system resets. Ultimately, this testing validates the overall system stability under dynamic transient loads.

3.5. Conclusion

The power architecture successfully resolves the electrical and spatial constraints of the buoy. The 2S topology provides a low quiescent baseline while safely managing dynamic load spikes. Mathematical modelling confirms the 2S12P design holds sufficient derated capacity to autonomously sustain the platform through a 105-day Antarctic winter. To achieve multi-season autonomy, this storage architecture integrates a fractional V_{OC} MPPT solar harvesting system, utilising monocrystalline cells and hardware-enforced NTC thermal limits to safely replenish the cells during the polar summer daylight [60].

4. Turbidity Sensor

4.1. Introduction

Turbidity, measured in Nephelometric Turbidity Units (NTU), was included on the buoy as it is a key indicator of water clarity caused by suspended particulate matter. Its primary climatic influence operates through albedo; suspended particles absorb rather than reflect incoming solar radiation, directly elevating sea surface temperatures and accelerating glacial melt [61]. Rising sea surface temperatures further strengthen thermal stratification, cutting off phytoplankton from nutrient-rich deep water [62], whilst warmer water accelerates bacterial breakdown of sinking organic matter, returning carbon to the water column rather than sequestering it in the deep ocean [63]. As a result, turbidity does not merely reflect the consequences of warming poles, it actively accelerates them. To capture this parameter autonomously, a turbidity sensor was developed and integrated into the buoy, shown as label I in Figure 2.

4.2. Literature Review

There are several recognised methods and standards for turbidity measurements. These primarily are optical approaches that use the change in reflective properties, however there are other methods including acoustic and gravimetric [64]. Table 6 below shows a summary of these methods and their pros and cons.

Table 6 - Turbidity measurement methods comparison

Method	Description	Pros	Cons
Nephelometric ISO7027 [67]	An optical scattering method in which the detector is positioned at 90° to the incident beam. ISO 7027 specifies an 860nm infrared(IR) source, to minimise absorption from dissolved organic matter and coloured substances.	High sensitivity at low NTU values. Colour and dissolved organic interference minimised by NIR source. Well standardised and internationally recognised (ISO 7027).	Saturates at high turbidity, limiting dynamic range. The requirement for the detector to be at 90 degrees makes the practical design more complex to package
180-degree Optical Attenuation [TURB6]	An optical method in which the source and detector are co-located on the same side of the sample, measuring light returned towards the source.	Source and detector can be mounted on a single face. Suitable for high-turbidity environments.	Less standardised than nephelometry. More susceptible to surface fouling on the optical window, which directly degrades signal quality.

Acoustic [64]	A method based on the attenuation and backscattering of ultrasonic waves by suspended particles.	Effective at high suspended sediment concentrations where optical methods saturate. Insensitive to sample colour and dissolved organics. Not subject to biofouling of an optical window.	Not directly mappable to the NTU scale, instead measure of total suspended solids. Poor accuracy
Gravimetric analysis [64]	A laboratory method in which a known sample volume is filtered and the retained solids dried and weighed to determine total suspended solids (TSS) concentration.	Provides an absolute, direct measure of suspended solid mass concentration. Not subject to optical or acoustic interference effects.	Destructive and non-continuous. Labour intensive and slow to execute. The relationship between TSS mass and optical turbidity is non-linear and particle-size dependent.

4.3. Design and Implementation

4.3.1. Fundamental Principle

A 180° optical attenuation configuration was selected for the buoy's turbidity sensor, primarily as this geometry allows the source and detector to be co-located on a single face, enabling the sensor to be mounted on the static end cap of the buoy without requiring a protruding component. This is particularly important in the context of polar seawater, which undergoes approximately 10% volumetric expansion upon freezing which could cause mechanical failure [65]. This configuration has been shown to be less sensitive compared to backscatter configurations like ISO7027 however it mitigates the risks of mechanical failure.

A challenge for any optical turbidity sensor deployed in an open environment is the influence of ambient light, which introduces a time-varying offset to the photodetector signal, and the variable optical response of different suspended particulate types. The sensors design addresses both through two complementary measures; the use of infrared excitation and detection, and firmware-based ambient light cancellation.

Infrared excitation and matched IR detection inherently rejects much of the visible ambient light spectrum, and the subsequently sea water attenuates the remained solar IR light [66]. Additionally, this reduces the effects of organic suspended matter that can be colour specific absorption behaviour [67].

Software-based ambient light cancellation supplements this by acquiring two measurement sets: an excited set, in which the IR source is active, and a non-excited set, in which it is not. Each set is averaged to produce a mean excited response and a mean non-excited response. Subtracting these isolates the backscattered component attributable to the IR source alone, removing the ambient light contribution from the final turbidity measurement. Therefore, by subtraction the of mean of the non-excited measurement from the excited measurements a purely backscatter dependent is found, as illustrated below in equations 4.1 and 4.2.

$$\overline{V_{Compensated}} = \overline{V_{Excited}} + \overline{V_{Non Excited}} \quad (Equation 4.1)$$

$$\therefore \overline{V_{Compensated}} = V_{Ambient Ligh} + V_{Back Scattered} - V_{Ambient Ligh} = V_{Back Scattered} \quad (Equation 4.2)$$

4.3.2. Electronic Design

The turbidity sensor has been implemented on a single PCB mounted to the static end cap of the buoy (shown in Fig 2, label 'I'). The schematic is shown in Figure 21:

Turbidity Sensor Schematic

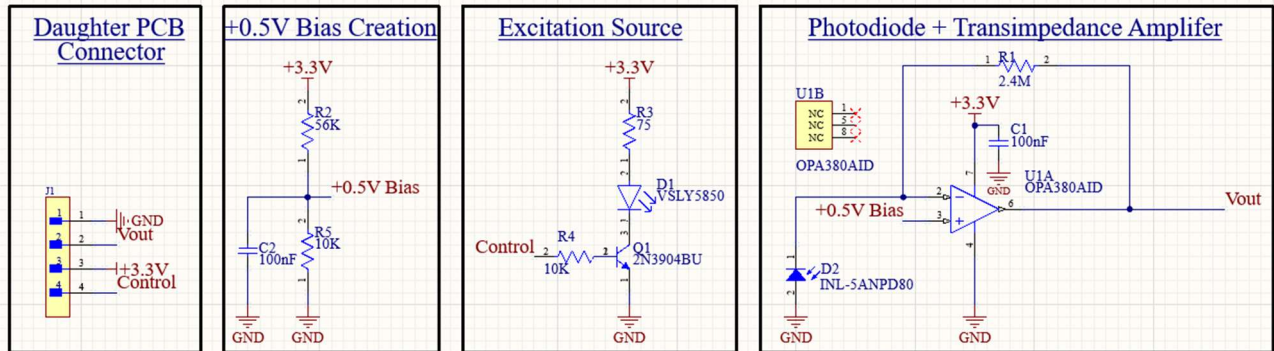


Figure 21 – Turbidity Sensor Schematic

A VSLY5850 LED was selected as the excitation source due to its 850nm peak emission wavelength, aligning with the ISO 7027 turbidity specification and matching the peak spectral sensitivity of the INL-5ANPD80 photodiode detector. The LED is switched via a 2N3904 NPN transistor, allowing the microcontroller to control excitation. The photodiode is supplied in a 5mm through-hole package with a black epoxy lens, providing an additional layer of optical filtering at the component level to reject out-of-band ambient light before the signal reaches the amplification stage.

Signal conditioning is performed by an OPA380AID transimpedance amplifier, selected for its low input bias current and single-supply, 3.3V compatibility. To prevent non-linear operation near ground the non-inverting input is biased to a +0.5V reference, shifting the output into the linear operating region. The transimpedance gain was determined analytically and refined experimentally to 300,000, chosen to utilise the range of the STM32's ADC whilst keeping the OPA380's feedback loop stable.

The PCB layout was designed to minimise board area, making use of 0805 surface mount components throughout. Beyond the space saving, the use of surface mount packages reduces stray capacitance at the TIA inputs reducing the risk of oscillatory behaviour in output. The resulting board footprint is 8cm². The mounting arrangement is shown in Figure 37 Section 10; the board is fixed to the end cap using stainless steel fasteners, positioning the LED and photodiode 15mm below the centre of the end cap to ensure the optical elements remain submerged at the buoy's natural waterline in its operational buoyancy position.

4.4. Testing

4.4.1. Methodology

Calibration was conducted using kaolin clay dissolved in 7 litres of distilled water, selected for its approximately linear relationship between mg/L concentration and NTU [68]. Kaolin was added in increments of approximately 0.5 g, spanning 0 g to 7 g, corresponding to 0–1000 mg/L and an approximate NTU range of 0–400 NTU. This upper bound was justified by Roberts et al. (2017), who recorded a peak of 250 NTU in a High Arctic lake following permafrost disturbance [69]. The sensor was mounted above the solution with its LED and photodiode positioned below the water surface. The kaolin was measured on a calibrated digital scale (± 0.005 g) and fully dispersed by stirring before each set of measurements. Five backscatter voltage readings were taken and averaged at each concentration to reduce electrical noise.

4.4.2. Results

As shown in Figure 22, an observable trend is present between the backscatter voltage measured and the NTU, consistent with Beer-Lambert attenuation of the IR source through the suspended medium [70]. Using the exponential line of best fit the relationship was characterised. This characteristic equation is shown in

Figure 22 linking the backscatter voltage and NTU, this was then introduced into the firmware such that a function returns a NTU value rather than simply the backscatter voltage.

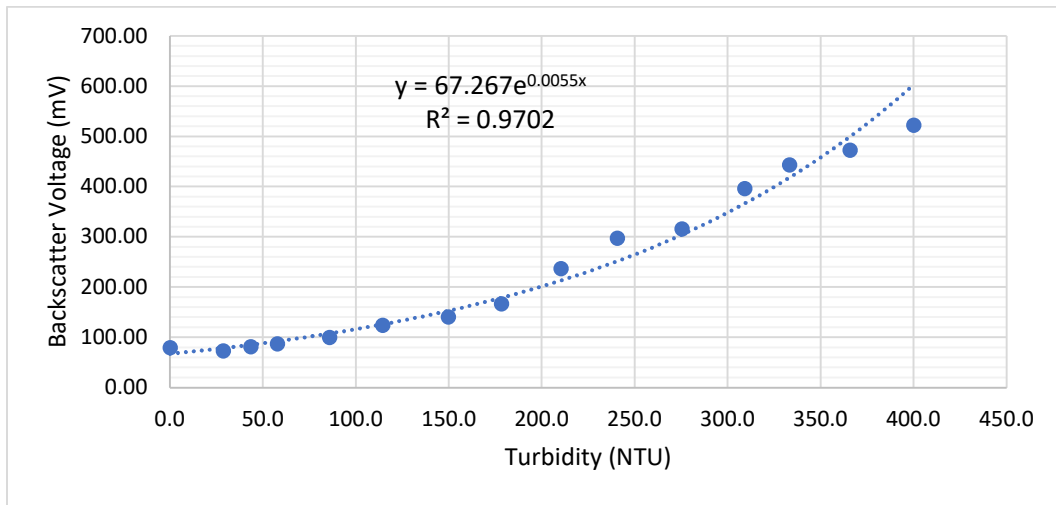


Figure 22 - Backscatter Voltage vs Turbidity NTU

4.5. Conclusion

The turbidity sensor was successfully calibrated across a range of 0-400 NTU, with the exponential characteristic equation implemented directly in firmware to return calibrated NTU values. The R^2 value of 0.9702 demonstrates a strong fit, validating the sensor's response as consistent with Beer-Lambert attenuation theory [70]. A key limitation is the reduced sensitivity at low turbidity values (0-50 NTU), where backscatter voltages cluster between 70-90 mV, constraining the sensor's ability to resolve fine differences in near-pristine open Arctic water which typically sits below 6 NTU [69]. The sensor is therefore best suited to detecting periods of elevated turbidity caused by glacial melt and sediment disturbance, which are the most climatically significant turbidity events in Arctic regions.

5. Salinity Sensor

5.1. Introduction

Salinity is recognised as an ECV due to its role in regulating ocean dynamics and climate processes [71]. Together with temperature, salinity determines seawater density and drives thermohaline circulation, which redistributes heat globally and moderates regional climates [72], [73]. Variations in salinity disrupts circulation and influence sea ice formation and stability. This is significant because sea ice has a high albedo, reflecting solar radiation and helping regulate global temperatures. Reduced ice cover increases ocean heat absorption, reinforcing warming through a positive feedback loop. These changes disrupt marine ecosystems, lead to food insecurity and coastal erosion [74], [75]. Accurate long-term data is critical for understanding and predicting climate behaviour and environmental impacts. To monitor this parameter autonomously, a salinity sensor was developed and integrated into the buoy, shown as label J in Figure 2.

5.2. Literature Review

There are several sensing configurations used for salinity measurement, most commonly, contact-based approaches being two and four-electrode systems. The two-electrode method is the simplest arrangement, where the same pair of electrodes is used to inject the excitation current into the seawater and measure the resulting voltage. Although AC excitation reduces electrode polarisation compared with DC excitation, the electrode-electrolyte interface still introduces double-layer capacitance. Subsequently, the response can become impedance-based rather than purely resistive [76]. These capacitive effects can be partially reduced using bipolar square-wave excitation and by sampling during the steady-state regions of the waveform [76].

The four-electrode method reduces the capacitive effects by separating current injection and voltage measurement. In this configuration, outer electrodes apply the excitation current, while inner electrodes measure the resulting potential difference. Since the voltage-sensing electrodes draw negligible current, the effect of electrode polarisation and capacitance on the measured voltage is significantly reduced. This improves measurement accuracy and stability compared with a two-electrode arrangement [77].

An alternative non-contact approach is inductive sensing. Taking advantage of pre-existing EMI SIT sensor as discussed in Section 2.2, where transmitter and receiver coils are used to infer conductivity from induced eddy currents. This avoids direct electrode contact with seawater, reducing corrosion and biofouling issues. However, inductive systems require more complex excitation and signal-conditioning circuitry, higher power consumption, and calibration due to their sensitivity to coil geometry, spacing, and positioning [78]. These limitations make inductive conductivity sensing less suitable for a low-power buoy platform. Another practical limitation is the high cost of commercially available salinity sensors, which can exceed £200 [79], [80]. Therefore, to reduce cost while maintaining suitable measurement performance, a custom four-electrode conductivity sensor using stainless steel electrodes was selected. This approach provides a balance between measurement accuracy, power consumption, cost, and system complexity. The four-electrode arrangement reduces capacitive measurement artefacts compared with a two-electrode design, while avoiding the increased complexity and power requirements of inductive sensing.

316L marine grade Stainless steel was selected as a cost-effective and well performing electrode material with strong corrosion resistance for prototype testing. Titanium electrodes coated with platinum or iridium oxide were also considered due to their improved corrosion resistance and reduced biofouling behaviour [81]. Despite this, stainless steel was considered appropriate for the project because it allowed a low-cost, sensor to be manufactured and experimentally calibrated within the projects scope.

5.3. Design and Implementation

The salinity sensor was located on the static side of the buoy, shown in Figure 2, label 'J', and interfaced with the STM32F411 rather than the STM32F405 used on the motherboard. This choice reduced cost and power consumption, as the F411 provided sufficient processing capability for the salinity subsystem. However, unlike the F405, the F411 does not include an onboard DAC. Therefore, a PWM-generated sine wave was passed through two Sallen-Key Butterworth low-pass filters to produce an AC sinusoidal excitation signal. This approach also allows future migration to a lower-power microcontroller while retaining the same analogue waveform generation circuitry.

The filtered sine wave was then passed through an AC coupling capacitor to remove its DC component from the STM output. Before exciting the outer two electrodes, the signal passed through a shunt resistor, as shown at the top of Figure 23. The voltage drop across this was used to determine the current. This differential voltage was passed into an instrumentation amplifier, shown at the bottom of Figure 24, which removed common-mode noise, applied gain, and added a 1.65 V offset. This offset centred the AC signal within the input range of the STM32 ADC, allowing the full ADC range to be used without clipping the negative half-cycle of the waveform.

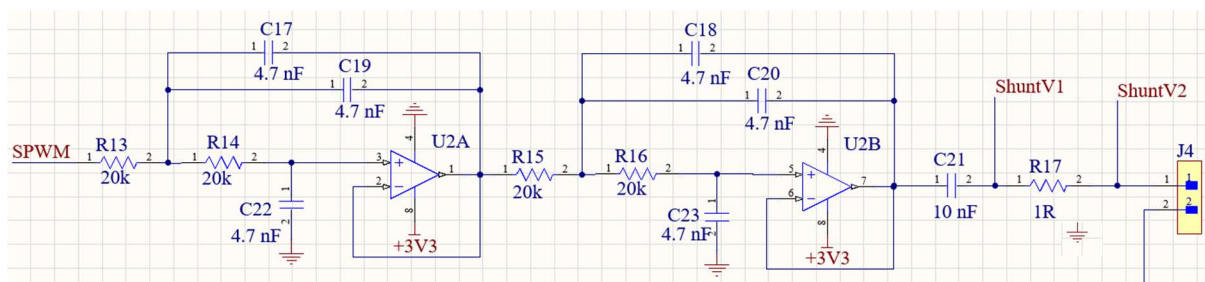


Figure 23 – Sallen-Key Circuit Schematic

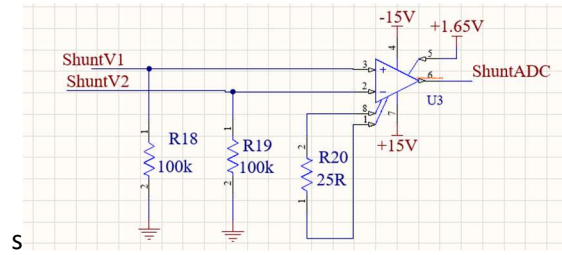


Figure 24 – Instrumentation circuit schematic

The voltage across the inner sensing electrodes was measured using a duplicated instrumentation amplifier circuit. This provided a second amplified and reference-shifted signal representing the potential difference across the seawater sample. Both the shunt voltage and sensing electrode voltage were then sampled by the onboard STM32 ADC. The RMS values of both signals were calculated in the STM32 software and were used to determine conductivity using equation 5.1, where K is the cell constant. An external temperature reading was also taken, allowing the measured conductivity to be corrected to a standard reference temperature before being used in a lookup table stored in flash memory to calculate salinity.

$$\sigma = K \frac{I}{V}, \text{ where } K = \frac{L}{A} \quad (\text{Equation 5.1})$$

5.4. Testing and Calibration

The salinity calibration test setup is shown in Figure 25. A measured volume of seven litres of deionised water was placed in the test container, with the four-electrode salinity sensor submerged in the solution. Salt was added in 25.00 gram increments up to a total mass of 300.00 g, producing a maximum salt concentration of 42.9 g/L. At each increment, the RMS voltage across the 1 Ω shunt resistor and the RMS voltage across the inner sensing electrodes were recorded using an oscilloscope. A calibrated EC210 conductivity meter was also used to record the reference conductivity and solution temperature.

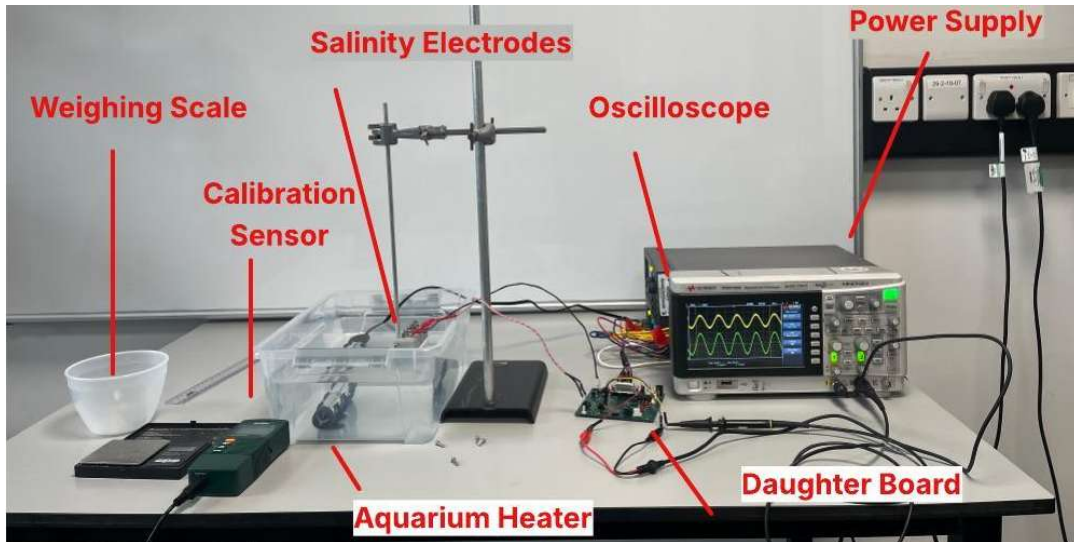


Figure 25 – Salinity calibration and testing setup

The measured conductance term, G , was calculated from the ratio of shunt voltage to inner electrode voltage. Since the shunt resistor was 1 Ω , the shunt voltage was directly proportional to the excitation current. The reference conductivity from the EC210 meter was then plotted against G , as shown in Figure 26. Linear calibration fits were obtained for measurements at 20.4°C and 25.5°C, giving gradients of 35.44 and 34.19, respectively. These gradients represent the effective calibration constant of the sensor, including the electrode geometry and gain scaling from the measurement circuitry.

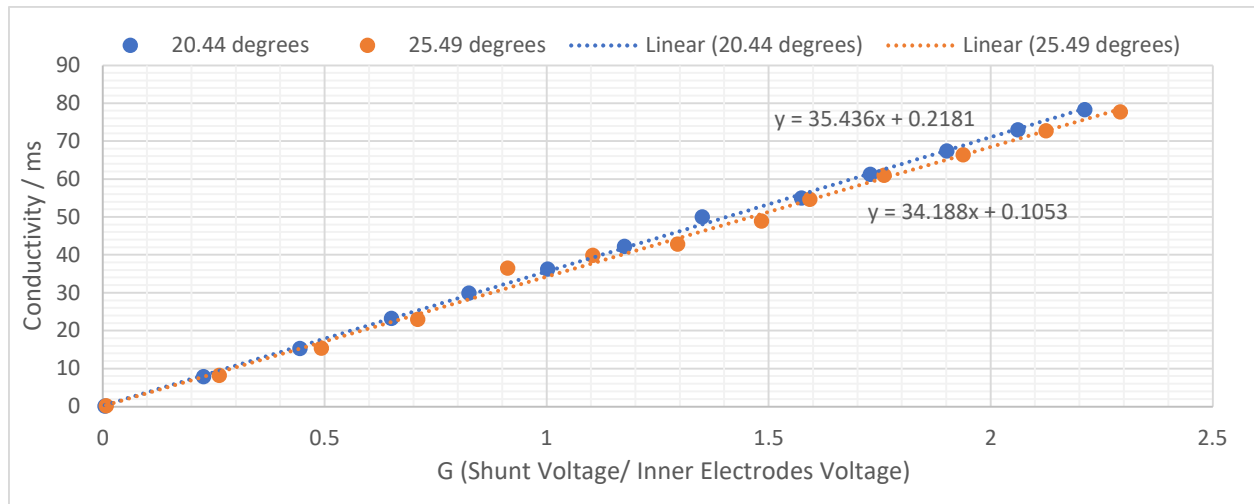


Figure 26 - Conductivity versus G

The calibrated sensor response was then used to calculate the conductivity at each salt concentration. This was plotted, as shown in Figure 27. The results show that conductivity increased approximately linearly with salt concentration, as expected given that adding salt increases the concentration of dissolved ions. The two temperature datasets show slight differences in gradient, indicating that temperature affects the conductivity response. However, the trend was not consistent enough to determine a reliable temperature compensation model. Future work involves testing over a wider and more controlled temperature range.

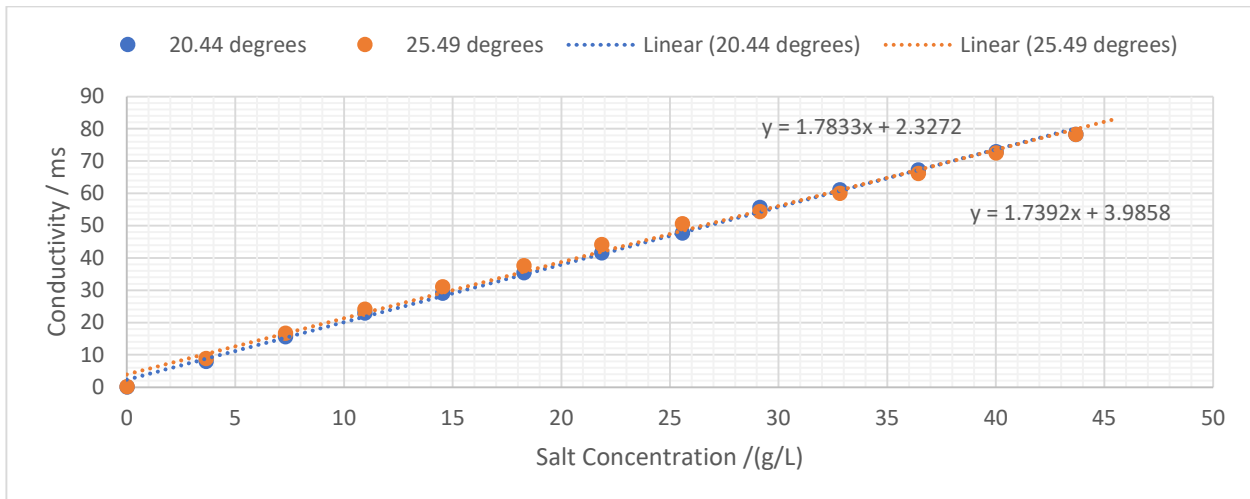


Figure 27 – Conductivity versus salt concentration

5.5. Conclusion

The four-electrode approach provided a simple and cost-effective method for measuring salinity while reducing the effects of electrode polarisation. By separating the current injection and voltage measurement electrodes, the design improved measurement stability compared with a basic two-electrode configuration. The use of a custom-built sensor also offered significant cost savings compared with commercial pre-calibrated salinity sensors, allowing more of the project budget to be allocated to other buoy subsystems.

6. Imaging System

6.1. Introduction

Imaging provides vital contextual information about the buoy's surrounding conditions, helping researchers interpret sensor measurements that may be influenced by environmental factors. Images can reveal sea ice coverage, weather conditions, icing, condensation, and other phenomena that may affect onboard sensors.

However, implementing an imaging system on a remote climate-monitoring buoy introduces significant challenges. Satellite communication is bandwidth-limited and expensive, at approximately £0.89-£1.09 per 1000 bytes [82], making transmission of raw image data impractical. Power consumption, limited microcontroller SRAM, and restricted processing capability further constrain the system. Therefore, a computationally efficient process is required to ensure only useful images are selected for transmission, minimising power usage and cost. To meet this need, a camera-based imaging system with onboard image quality filtering was developed and integrated into the buoy platform, shown as label K in Figure 2.

6.2. Literature Review

Image quality assessment methods fall into three broad categories: full-reference (FR) methods, which require a pristine reference image; statistical no-reference (NR) methods, which assess quality from pixel-level features; and learning-based blind Image Quality Assessment (IQA) methods such as Blind/Referenceless Image Spatial Quality Evaluator (BRISQUE), which use scene statistics models to produce a distortion-generic quality score [83],[84]. FR methods are immediately unsuitable as no reference image exists at capture time. Learning-based NR methods, while more accurate, require model training and carry memory overhead incompatible with a MCU operating under tight SRAM and constraints [84].

Statistical NR metrics are therefore the appropriate class for this application. Laplacian variance is among the best-validated sharpness operators in the NR IQA literature, identified as consistently high-performing due to its sensitivity to high-frequency content and low computational cost [85]. Histogram entropy captures information independently of mean brightness, while dark and saturated pixel ratios detect the two exposure failure modes most common in marine environments. Together, these metrics cover complementary failure modes without model inference. Thresholds are determined using the Otsu method (discussed in Section 6.4), providing a data-driven alternative to manual tuning [86].

6.3. Implementation

The system captures compressed JPEG images using a two-megapixel camera at QQVGA resolution (160×120). At this resolution, images are typically 2-5 KB, reducing transmission cost and memory usage. As processing is performed on the STM32F405RGT6 MCU, which lacks JPEG hardware acceleration, each image is decoded in software and converted into greyscale (~19.2 kB). Greyscale processing was selected to minimise computational complexity while retaining sufficient information for image quality assessment.

A set of computationally efficient image metrics (Table 7) is then calculated and compared against thresholds determined using the weighted Otsu method [86]. Images are rejected if they fail any threshold, allowing early elimination of underexposed images (high dark pixel ratio), overexposed/clipped images (high saturated pixel ratio), low-information images (low entropy or range) and blurred images (low Laplacian variance). Table 7 summarises the metrics used and their purpose.

Table 7 - Processing metric uses and purposes

Metric	Purpose
Laplacian Variance	Sharpness - stops blurry image
Histogram Entropy	Advance contrast e.g. spread of pixel distribution and intensity – Stops information less images
Range	Simple contrast e.g. spread of intensity – stops images of just one colour
Saturation (Pixel) ratio	Clipping – stop images with too many bright and too dark pixels
Dark Pixel ratio (DPR)	Stops too dark photos

Images that pass this stage proceed to a scoring stage, where a weighted score is calculated as shown in Equation 6.1:

$$\text{Score} = 0.5 \cdot \text{Laplacian Variance} + 0.3 \cdot \text{DPR} + 0.2 \cdot \text{Histogram Entropy} \quad (\text{Equation 6.1})$$

A final threshold is then applied to determine whether the image is suitable for transmission. This stage enables nuanced classification of borderline cases and provides a clear quantitative metric for user decision-making when requesting image retrieval. This two-stage approach balances computational efficiency with classification robustness, ensuring that only high-quality and informative images are transmitted while remaining compatible within the constraints of the embedded platform.

6.4. Threshold Tuning

To determine optimal thresholds for each image feature, the Otsu method was applied. Otsu thresholding identifies the value that minimises intra-class variance between two classes, effectively selecting the point that best separates their feature distributions.

A labelled dataset was constructed, consisting of 295 Arctic (good) images sourced from the National Oceanic and Atmospheric Administration (NOAA) Antarctic collection, and 300 non-Arctic (bad) images comprising well-exposed everyday scenes [87].

All images were pre-processed by resizing to QQVGA resolution (160×120 pixels) and converting to greyscale, matching the constraints of the embedded implementation. From each image, a range of statistical and structural features were extracted, including intensity-based metrics (mean, variance, dynamic range), exposure indicators (dark and bright pixel ratios, saturated pixel ratio), and quality measures (Laplacian variance, RMS contrast, and histogram entropy). Each feature was analysed by plotting class-wise histograms, and the Otsu method was applied to determine optimal thresholds for separating Arctic and non-Arctic images, as summarised in Table 8 below.

Table 8 - Optimal threshold values (Otsu Algorithm)

Feature	Threshold	Feature	Threshold
Mean	127.7	Laplacian Variance	4297
Variance	3210	RMS Contrast	0.455
Range	206	Saturated Pixel Ratio	0.088
Dark Pixel Ratio	0.198	Histogram Entropy	6.66
Bright Pixel Ratio	0.312		

6.4.1. Computation reduction via correlation analysis

A Pearson correlation analysis was performed across all feature combinations to identify redundancy. Strong correlations were observed between mean intensity, RMS contrast, and dark pixel ratio, as well as between variance and RMS contrast, indicating that these features provided overlapping information.

Consequently, mean intensity, variance, RMS contrast, and bright pixel ratio were removed from the feature set to reduce computational complexity while maintaining classification performance. This reduction is particularly important for efficient execution on the STM32 microcontroller.

6.4.2. Weighting Optimisation for scoring model

A grid search was applied to the reduced feature set to find the combination of feature weights that maximised classification accuracy. This identified Laplacian variance (54%) and dark pixel ratio (46%) as the dominant features.

However, the dataset did not fully represent certain failure cases (e.g. overexposed or low-entropy images), leading to an underestimation of features such as entropy. To improve robustness and generalisation, a hybrid weighting was adopted. This retains the most discriminative features while incorporating entropy to account for exposure-related edge cases, resulting in Equation 6.1.

6.5. Conclusion

The tuning process was performed on a relatively small dataset consisting of arctic images and general everyday scenes. While sufficient for initial validation, this may not fully represent the variability of real-world conditions. In particular, non-arctic images with similar lighting characteristics to arctic scenes may reduce classification accuracy. A larger and more representative dataset, including both arctic and non-arctic environments under varied conditions, would improve parameter robustness and generalisation.

7. External Temperature Sensor

7.1. Introduction

This subsection covers the development and testing of a modified thermistor string present in label L of Figure 2, designed to provide temperature measurements across all orientations in both terrestrial and aquatic environments. Temperature sensing is a core function of existing climate-sensing buoys, as identified in Section 1.6.1, and is required for the completion of Objective 3 to support polar climate monitoring.

7.2. Literature Review

This section details the advantages and limitations existing temperature sensors in polar climates, all three of the options presented in Table 9 are utilised in fixed ground stations as well as mobile buoy solutions.

Table 9 - Advantages and limitations of temperature sensing solutions

Device/Method	Description	Advantages	Limitations
Thermocouple	Generated voltage at a junction of two different metals produces voltage and temperature difference between measurement and reference [88].	Highest range of all methods -200 °C to 2000+ °C, low relative cost, quick response time, low relative cost, simple implementation [89]	Low accuracy ($\pm 1^\circ\text{C}$), cold-junction compensation and a high-gain circuit required for the low signal level, operation range suited for high temperature, unsuitable for polar deployment specification.
Negative Temperature Coefficient (NTC) Thermistor Chains	A passive component that exhibits a decrease in resistance as temperature increases. Thermistor strings mount chain of NTC thermistors with shared ground/power source.	Extensive usage in polar conditions on buoys such as IMB and SIMB, extremely scalable multiplexing available, higher accuracy compared to thermocouple and PRT [90].	Smaller operating range (-80 °C to 150 °C), non-linear, requires additional firmware calibration, self-heating requires low current source, mechanically more complex [91].
Platinum Resistance Thermometer (PRT)	Platinum based thermometer with near-linear resistivity change (0.1% to 0.6% per °C) [92].	Highest accuracy and resolution (0.01 °C), simple operating principle, maintain calibration over extended time, wide operating range (-200 °C to 600 °C) [93].	Relatively expensive compared to other options, low signal requires high resolution ADC increasing circuit complexity, resolution too high for specification range.

Based on this analysis, thermocouples can be excluded on poor accuracy and a range biased toward high-temperature applications, neither of which is suitable for a polar climate sensing application. PRTs offer

extremely high accuracy, but the implied cost of a precision external ADC and the additional analogue circuit complexity would be too expensive and infeasible for the proposed buoy design, which uses the built-in 12-bit ADC on the STM32F411RET6. The NTC thermistor chain is the most suitable choice, having been established in existing polar buoys referenced in Section 1.6 including the IMB and SIMB, it is an extremely low-cost option (<£2 per unit) [94] with sufficient accuracy for the use case, and the mechanical complexity limitation can be compensated by the design choices set out in Section 7.3.1.

7.3. Design and Implementation

7.3.1. Temperature Probe Design

Several temperature harness configurations were prototyped during the design process, including a conventional loose-hanging thermistor string arrangement. However, this configuration was deemed unsuitable, as the trailing cable interfered with both the buoy's mobile rotating operation and the surrounding natural environment.

The final design adopted a ring configuration of thermistors mounted around the buoy's inner tube, as shown in Figure 28. Based on the operating principle of the thermistor string described in [95], this approach replaces the trailing wire arrangement with external fixed bolt points distributed around the buoy. The resulting quad-channel design ensures reliable contact with the surface medium regardless of the buoy's orientation.

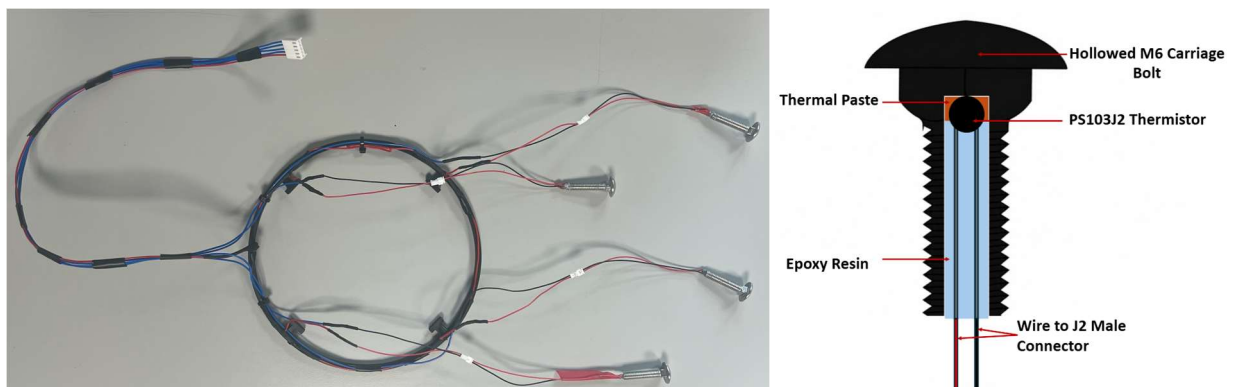


Figure 28 - Temperature sensor harness design and cross section of probe

The mechanical assembly uses the PS103J2 thermistor mounted within a hollowed stainless steel M6 coach bolt, as shown in Figure 28. Thermal paste was applied between the sensor tip and bolt interior to maximise thermal transfer, while the sensor legs were fixed with epoxy resin to provide structural support and additional waterproofing. The coach bolt operates as a thermal transfer medium and protective waterproof housing while the integration of J2 male connectors at the thermistor pins allow sensors to be individually hot-swapped for improved serviceability.

7.3.2. Sensor Performance Justification

The PS103J2 is a precision small-form-factor NTC thermistor, selected for its size and broad operating range of $-80\text{ }^{\circ}\text{C}$ to $+150\text{ }^{\circ}\text{C}$, which comfortably exceeds the $-40\text{ }^{\circ}\text{C}$ arctic climate requirement outlined in Specification 5. The sensor maintains an accuracy of $\pm 0.1\text{ }^{\circ}\text{C}$ above $0\text{ }^{\circ}\text{C}$ without calibration [94]; with calibration this figure is achievable at lower temperatures, as demonstrated in Section 7.4. The sensor exhibits an intrinsic J-curve temperature coefficient of $-4.4\text{ }^{\circ}\text{C}/^{\circ}\text{C}$ with a β value of 3892 K, producing a smooth response profile that simplifies firmware calibration and modelling.

7.3.3. Implementation

Each thermistor is wired as shown in Figure 29. One leg of the PS103J2 is connected to a reference 15 k Ω pull-up resistor, and the other leg shares a 3.3 V VCC rail with all four thermistors, creating a voltage divider circuit described by equation 7.1:

$$V_{OUT} = \frac{V_{CC} \cdot R_{NTC}}{(R_{REF} + R_{NTC})} \quad (\text{Equation 7.1})$$

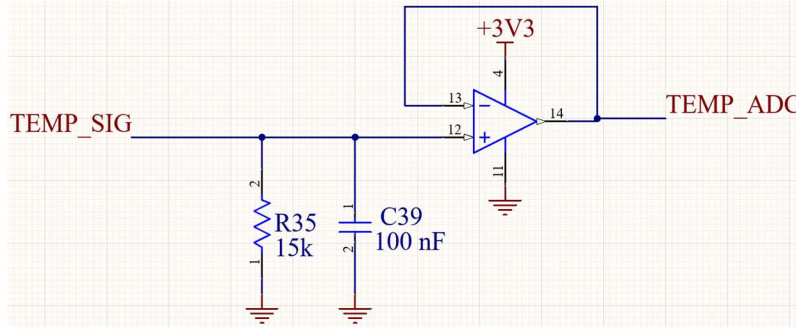


Figure 29 - Per channel thermistor schematic

R_{ref} controls the temperature of peak sensitivity with the highest sensitivity when R_{ref} is equal to R_{ntc} , which sits at 10 °C in the present configuration calculated using the differential of the voltage divider equation shown in equation 7.2. For live demonstration purposes the peak sensitivity has been increased beyond the typical polar deployment region to allow testing from 35 °C down to -5 °C; in the final polar configuration this resistor value would be reduced. Reference resistor tolerances are 0.1%, providing 6% improved accuracy when compared to 1% tolerance resistors for a negligible cost increase.

$$\frac{dV_{OUT}}{dT} = \frac{-\beta \cdot V_{CC} \cdot R_{REF} \cdot R_{NTC}}{(T^2 \cdot (R_{REF} + R_{NTC})^2)} \quad (\text{Equation 7.2})$$

The output of each divider feeds one channel of an LM324A quad amplifier configured as a unity-gain voltage follower. This package supports all four sensors using a single IC including two power pins, which reduces overall component count and PCB footprint compared to four discrete buffer implementations, while allowing for a shared decoupling network. The resulting resistance, R , calculated using the V_{out} is processed through the Steinhart-Hart equation shown in equation 7.3 to retrieve the temperature in °C where A , B and C are fixed coefficients

$$T(^{\circ}C) = \frac{1}{(A + B * \ln(R) + C * (\ln(R))^3)} - 273.15 \quad (\text{Equation 7.3})$$

The LM324A's 125 nA worst-case bias current [97] across the divider source impedance gives a 0.42 °C offset at the -40 °C sensitivity of 8.6 mV/°C, and its 0-1.8 V common-mode range is exceeded at -40 °C causing distortion. This is acceptable for ~ 0 °C demonstration but would be replaced in the final build by the TLV9064, providing no input cutoff and a significantly lower ~ 0.5 pA CMOS bias current removing both error sources.

7.4. Testing

The assembled string was submerged in an ice bath and tested through nine set temperatures spanning -5 °C to +35 °C alongside a calibrated reference thermometer. ADC readings were converted to temperature using the Steinhart-Hart equation defined in Section 7.3.3, where an average of five readings was taken and compared against the reference shown in Figure 30.

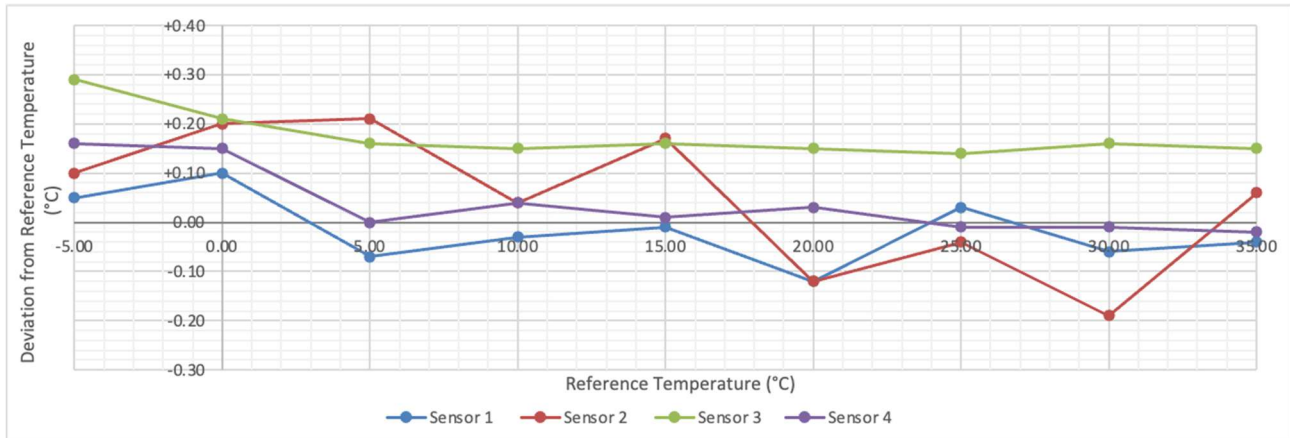


Figure 30 – Measured temperature error versus reference temperature

The peak deviation was 0.29 °C, recorded on Sensor 3 at a temperature of –5 °C. Deviations grew on all thermistor channels below 5 °C, with most readings at the two coldest setpoints sitting outside the ± 0.1 °C of the calibrated temperature. This behaviour matches the LM324A input common-mode limitation identified in Section 7.3.3 and is removed in the final build by substituting the TLV9064 amplifier. Across the 5 °C to 35 °C range Sensor 4 held within ± 0.04 °C, while the other three channels showed per-sensor offsets most likely related to mechanical impurities, resistor tolerances and LM324A channel matching [96].

7.5. Conclusion

The external temperature subsystem successfully integrates four PS103J2 NTC thermistors into a fixed-point ring around the buoy, providing reliable waterproofing, rapid thermal transfer, and hot-swappable modularity through the coach-bolt and J2 connector design. This section achieves all targets outlined in Specification 2 with an accuracy of ± 0.29 °C. The component choices exceed the –40 °C specification requirement and provide sufficient accuracy after calibration despite limitations, as validated by the testing in Section 7.4. Substituting the TLV9064 amplifier in the finished polar build will increase the input common mode range to a lower temperature, allowing for successful polar data acquisition.

8. GUI and Communications

8.1. Introduction

Polar climate monitoring buoys should be able to transmit climate and oceanographic data to researchers remotely without manual intervention. The communications subsystem provides an interface between the buoy and the ground station, enabling data transmission. In real polar deployment, autonomous buoys require satellite communications link that offer global coverage. The Iridium 9603N Short Burst Data (SBD) service is chosen as the target technology due to its pole-to-pole satellite coverage, support for two-way communication and high-power efficiency [98]. However, for the purposes of this project, the use of Iridium hardware presents practical constraints. The Iridium 9603 module costs £150 with each SBD message incurring an airtime charge of approximately £1.5 per 1000 bytes, without subscription fees [99]. Hence, Bluetooth Low Energy (BLE) using a HM-10 module is implemented as a communications channel to emulate Iridium under conditions more restrictive than the intended satellite connection [100]. The ground station software is presented in the form of a Graphical User Interface (GUI) implemented in Python using the Tkinter framework. The GUI is designed to transmit sensor selection commands to the buoy, receive and reconstruct data packets, store readings in CSV logs and display sensor data. This section covers the communication design choices, protocol implementation, testing and subsystem limitations with respect to the target satellite link.

8.2. Literature Review

Several satellite systems were evaluated against the polar coverage and power requirements set for this application. Inmarsat and Globalstar do not provide coverage for extreme polar regions [101]. Starlink provides arctic coverage but draws 25-40 W at its terminal, which is incompatible with the power budget of this application [102]. ARGOS, widely used for older oceanographic drifters, supports only one-way uplink transmission with latencies of up to two hours [103]. Iridium SBD's adoption in existing polar platforms such as the OpenMetBuoy and Datawell Waverider confirms its suitability for this application [104], [105].

Several low data rate technologies were considered as prototype substitutes for the Iridium link but BLE was chosen due to its low cost and simplicity. The HM-10 module provides a simple UART AT-command interface similar to Iridium 9603N, allowing the same firmware abstraction with minimal changes [100]. This requires the implementation of packet fragmentation and reassembly in software. The total payload transmitted for all defined sensors, excluding camera images fits within a single Iridium Mobile Originated (MO) message. However, transmitting compressed JPEG images (2-5 kB) over Iridium would require multiple SBD sessions and this is a real deployment constraint that is not tested in this prototype.

8.3. Design and Implementation

The Iridium SBD protocol supports MO messages up to 340 bytes and Mobile Terminated (MT) messages up to 270 bytes per session [98]. By comparison, the HM-10 module is limited to a 20-byte Attribute Protocol Maximum Transmission Unit (ATT MTU) [100]. Two message types are defined where the structure is compatible with Iridium. These structures are illustrated in Figure 31.

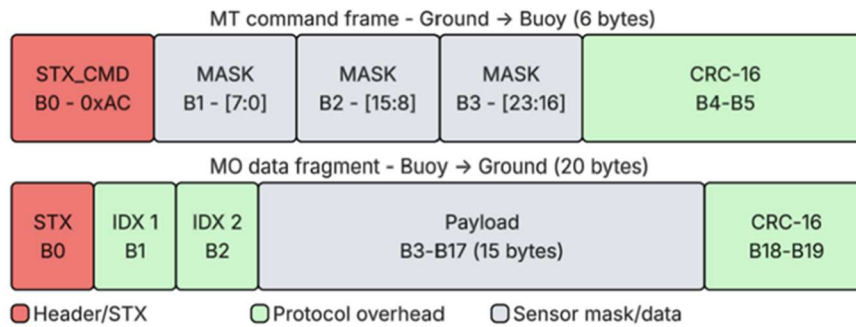


Figure 31 – Protocol implemented with BLE

The MT command frame (ground to buoy, 6 bytes) consists of the following: 1-byte header (STX_CMD, 0xAC), a 3-byte sensor selection bitmask and 2 bytes for a CRC-16 checksum. Each of the 24 bitmask bits corresponds to a specific sensor data type. This allows the operator to request any combination of measurements in a single transmission. The complete frame fits within the Iridium MT 270-byte limit [98].

The MO data fragment (buoy to ground, 20 bytes) consists of the following: 1-byte header encoding both data classes (sensor or image) and whether the fragment is the final one in the message, a 2-byte fragment index, 15 bytes of application payload, with the remaining 2 bytes for a CRC-16 checksum. CRC-16 is used for error detection and is also implemented on Iridium systems. Following successful reassembly of a complete MO message, the ground station transmits a single-byte acknowledgement to the buoy.

The HM-10 module is configured as a Generic Attribute Profile (GATT) peripheral with a single characteristic (UUID 0000FFE1) used for both sending and receiving data [100]. It communicates with the STM32 microcontroller through a 3.3 V UART connection and is located on designator R in Figure 2.

8.4. Testing (GUI)

The ground station GUI consists of three main software components in Python. The communications component manages BLE scanning using the Bleak library, GATT connections, command transmissions and

reassembly of fragmented BLE packets received from the HM-10 module before decoding them as either sensor or image data. The data management component stores received sensor readings in a structured CSV log with timestamps and session identifiers, which also serves as an interface between the communications and display layers. Figure 32 shows the Tkinter-based GUI, developed to validate BLE communication, displaying live sensor telemetry from the buoy. It confirms the system can successfully transmit sensor readings and respond to data requests from a host device.

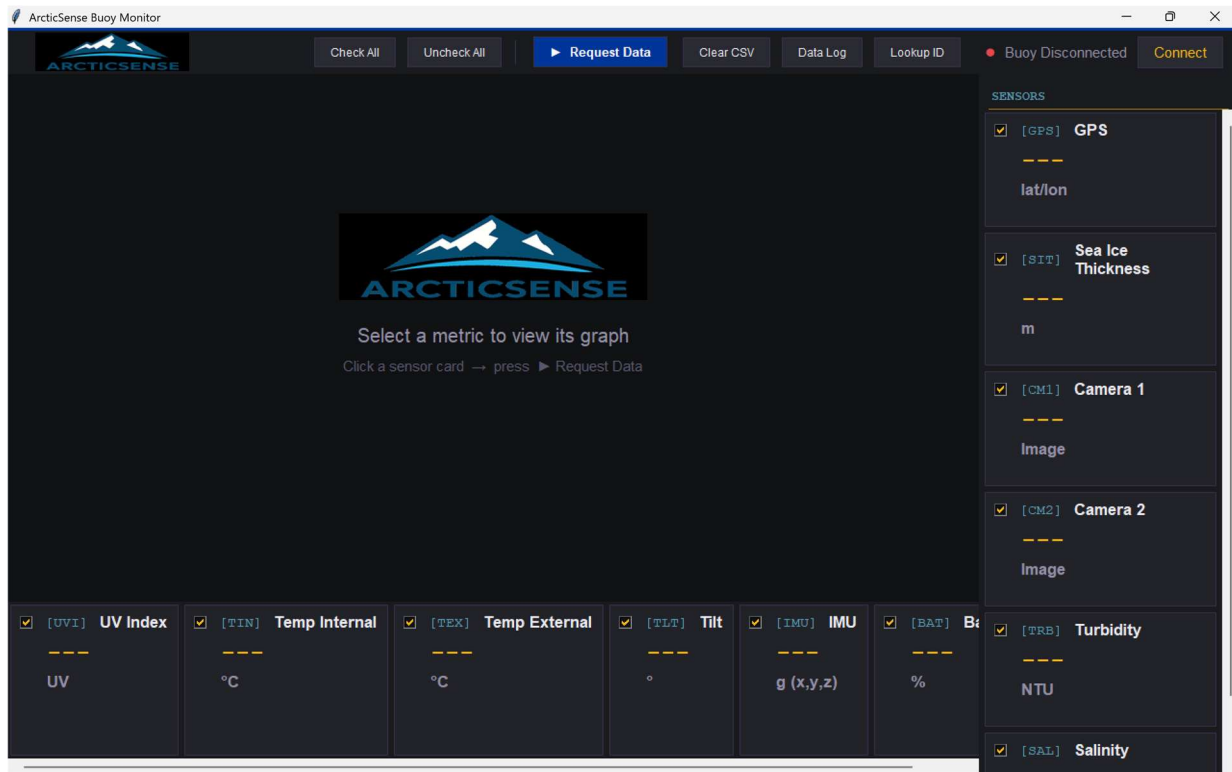


Figure 32 – Buoy Monitor GUI application

8.5. Conclusion

A BLE-based communications link and Python ground station GUI have been designed and validated as a prototype for the communications subsystem on the buoy. The bitmask command structure, sensor encoding and ground station architecture is validated against the more restrictive 20-byte BLE constraint and are directly compatible with an Iridium SBD implementation for all sensor data, excluding cameras.

9. IMU

9.1. Introduction

The inertial measurement unit (IMU) subsystem provides wave characterisation capability to the buoy, measuring the period, dominant frequency, and spectral distribution of surface waves at the deployment site. Wave spectra, which describes how wave energy is distributed across different frequencies, is a primary parameter in physical oceanography, used in storm forecasting, offshore energy assessment, and polar climate monitoring [106]. Accurate wave data is particularly relevant in polar environments where sea state directly influences ice formation, drift patterns, and the structural loading experienced by autonomous platforms during extended deployment [106]. The subsystem uses a single MPU6050 six-axis MEMS IMU, mounted internally within the buoy, which measures the angular motion of the buoy as it responds to passing waves [107]. By applying a Fast Fourier Transform (FFT) to the derived roll angle signal, the dominant wave frequency, period, and a 16-bin power spectrum are extracted autonomously with no external reference [108]. The low cost and low power draw of the MPU6050 make it well suited to the long-

duration deployment requirements set out in Specification 3 (Section 1.5). The IMU is mounted on the auxiliary daughter board, both of which are integrated into the buoy platform and shown as label M in Figure 2.

9.2. Literature Review

The Datawell Waverider is the international standard reference buoy for wave measurement, deriving wave height and period by double-integrating a triaxial accelerometer signal and applying spectral analysis [8]. It achieves centimetre-level accuracy and is endorsed by the WMO for operational wave monitoring networks. However, at over £25,000 per unit with requirements for permanent mooring and satellite telemetry, it is entirely unsuitable for a low-cost buoy.

The low-cost MEMS IMU based wave buoy achieves wave statistical parameters within 10% of co-located Datawell reference measurements while avoiding low-frequency drift, shown in [109]. This approach also yields a full power spectrum through FFT processing, which is more valuable than simpler time-domain methods such as zero-crossing detection, which gives only a single dominant period estimate. In polar environments, the dominant wave types are short-period wind-driven waves and ocean chop, typically falling within the 0.5 to 5 Hz frequency range, rather than the long-period swell of open ocean environments which is attenuated by sea ice cover [106],[109]. The full wave spectrum is therefore particularly useful in this context as it enables identification of multiple simultaneous wave components which influence ice break-up dynamics and buoy structural loading during Arctic deployment [106]. These characteristics informed the selection of a 0.3125 Hz bin resolution and 5 Hz upper frequency limit for this implementation, with detection of longer period swell identified as a direction for future work.

9.3. Testing

Testing was conducted in two stages: first a synthetic signal injection test to validate the FFT processing chain independent of hardware, followed by a physical pendulum test to validate the sensor's ability to capture real motion.

To verify the FFT pipeline, a 0.625 Hz sine wave was injected directly into the 32-point wave sample buffer, bypassing the physical sensor entirely. This frequency corresponds to bin 2 of the FFT output at the configured 10 Hz sample rate, where each bin represents a discrete frequency interval of 0.3125 Hz derived from the ratio of sample rate to FFT size. Table 10 confirms the FFT pipeline is correct and capable of accurately resolving a known frequency component from the roll angle signal.

Table 10 - Synthetic test expected vs observed results

Parameter	Expected	Observed
Dominant bin	2	2
Dominant frequency (Hz)	0.625	0.625
Dominant period (s)	1.6	1.6

To validate the sensor's response to real physical motion, the MPU6050 was mounted on a 25 cm pendulum using a weighted block as a counterweight to maximise pendulum momentum and reduce the effect of jumper cable drag. Based on the simple pendulum period formula:

$$T = 2\pi \sqrt{\frac{L}{g}} \quad (\text{Equation 9.1})$$

A 25 cm pendulum has a theoretical natural period of approximately 1.0 second, corresponding to a frequency of 1.0 Hz and an expected dominant bin of 3 at the configured 10 Hz sample rate and 32-point

FFT. As discussed in the literature review, this frequency range is appropriate for the polar deployment context of the buoy platform.

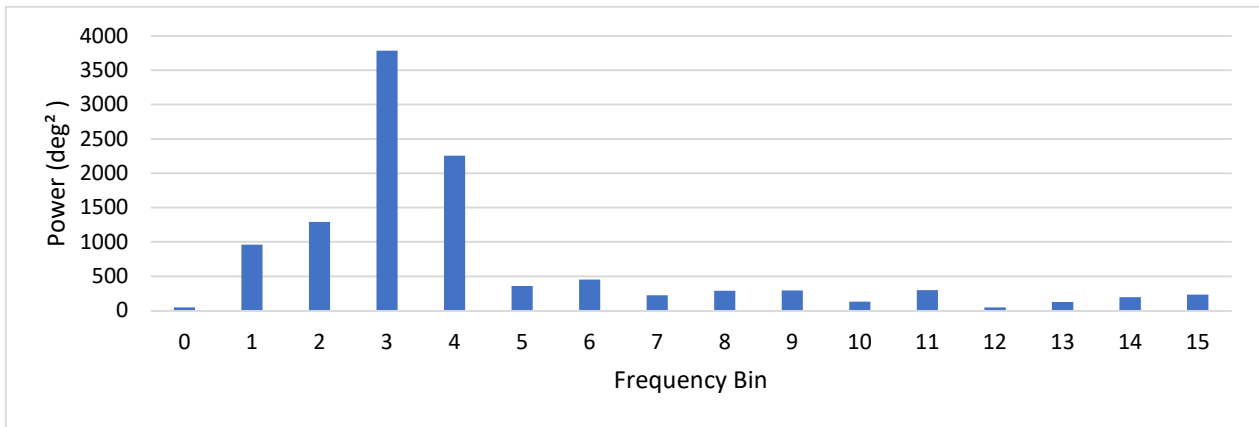


Figure 33 - IMU Pendulum test, Wave Spectra chart

The results, in Figure 33, show a clear dominant peak at bin 3, corresponding to a frequency of 0.9375 Hz and a period of 1.067 seconds. This is in close agreement with the theoretical natural period of 1.004 seconds predicted for a 25 cm pendulum by the simple pendulum formula, representing an error of approximately 6%. The remaining bins show significantly lower energy, confirming the FFT is correctly isolating the primary motion frequency. The small discrepancy between the measured and theoretical period is consistent with the damping effect of the jumper wire connections introducing a slight retarding force on the pendulum swing.

9.4. Conclusion

The IMU subsystem successfully demonstrates wave period and spectral extraction using a low-cost MPU6050 MEMS sensor and FFT-based signal processing. Synthetic validation confirmed the FFT is mathematically correct, and physical pendulum testing produced a dominant frequency of 0.9375 Hz against a theoretical prediction of 0.996 Hz, representing a 6% error consistent with minor experimental damping. The subsystem meets the low-cost and low-power requirements of the project and provides a foundation for continuous autonomous wave characterisation during polar deployment.

10. Mechanical Design

10.1. Introduction

The buoy's mechanical design, as specified in Section 1.5, must conform to the dimensions of a NATO sonobuoy tube to allow aerial deployment using existing infrastructure. However, given the sensing system equipped on the buoy, the mechanical design is not simply a housing; instead, it is critical to sensor operation and subsystem performance. Nonetheless, implementing a fully deployable and production-ready design is beyond the scope of this project. Consequently, the design makes significant use of 3D-printed components to facilitate a full-system demonstration within the project's limited timeframe.

As demonstrated in Section 2.3.1, the orientation of the EMI coils has a direct relationship with the magnitude of eddy currents induced within the seawater and, subsequently, the receive coils' ability to measure them. Therefore a mechanism is required within the buoy to ensure the coils remain in a HCP configuration relative to the seawater target throughout autonomous deployment.

To guide the mechanical development a list of design requirements were agreed, as shown below:

- DR1: Meet NATO sonobuoy standard for overall buoy dimensions.
- DR2: Passively self-level the EMI coils to ensure they remain in an HCP configuration.

- DR3: Provide sufficient capacity for peripheral sensors, batteries, and solar panels.
- DR4: Ensure all orientation-sensitive components, such as the cameras, UV sensor, and solar array, are appropriately positioned to maximise their performance.
- DR5: Provide a design suitable for mass manufacturing and widespread deployment.
- DR6: Where possible, use non-conductive materials and, where this is not possible, use low-conductivity materials such as stainless-steel fasteners to minimise EMI interference.

The proposed mechanical design can be divided into four primary assemblies; these sections are annotated in Figure 34. The following section of this report will analyse the design and reasoning behind each.

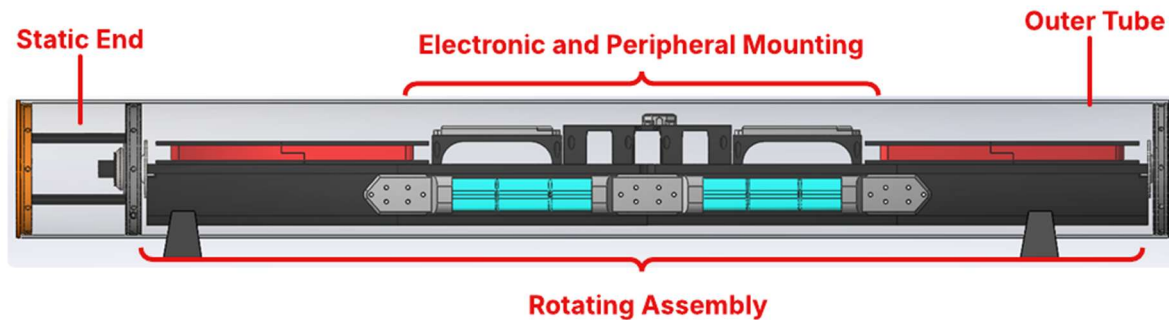


Figure 34 – Full mechanical test assembly

10.2. Literature Review

A review of NATO sonobuoy designs was carried out to identify common design trends and relevant information regarding their construction. However, given that most sonobuoys are designed for military applications, detailed design information is not readily available in the public domain.

As discussed previously, a key objective and design requirement for the buoy is compatibility with aerial deployment through a NATO sonobuoy tube. Several NATO-standard sonobuoy form factors exist, as shown in Table 11 below. Due to the requirement for the EMI coils to be as large as possible, combined with the need for substantial onboard energy storage to support extended deployments, the Class B format was selected for the buoy to be designed around, therefore in line with DR1.

Table 11 - Sonobuoy Tube comparisons, adapted from [110][111]

Sonobuoy Class	Length (mm)	Diameter (mm)	Maximum Weight (Kg)
A	914.40	123.82	17.69
B	1524.00	174.62	n/a
D	381.00	123.82	n/a
F	304.80	123.82	5.44
G	419.00	123.82	8.16

10.3. Design

All mechanical design work was completed in SolidWorks. The design process relied heavily on the use of shared design parameters across all part and assembly files, allowing for standardised dimensions and tolerances throughout the system. Additionally, several sub-assemblies and full test assemblies were completed in SolidWorks to verify component alignment and assembly prior to committing resources to 3D printing or machining parts. All CAD present in this report is available in the appendix.

Throughout the CAD process, the design requirements were continuously reviewed, particularly DR5, which specified that components should be suitable for mass manufacturing using conventional production methods such as injection moulding and CNC machining. Consequently, it is expected that the proposed design could be transitioned to manufacture with minimal design changes.

10.3.1. Rotating Assembly

At the centre of the mechanical design is a freely rotating I-beam. An I-beam cross-section was selected as it provides a large area moment of inertia while minimising the internal volume consumed. The I-beam consists of four unique 3D-printed sections, each 330 mm in length, bolted together using fishplates and dowels to maintain structural rigidity. The sections were designed with provisions for M5 and M3 stainless steel heat-set inserts, allowing additional components, brackets, and sensors to be mounted securely in a serviceable manner. Figure 35 below shows the complete rotating assembly model produced prior to printing, highlighting the key components within the assembly and placement.

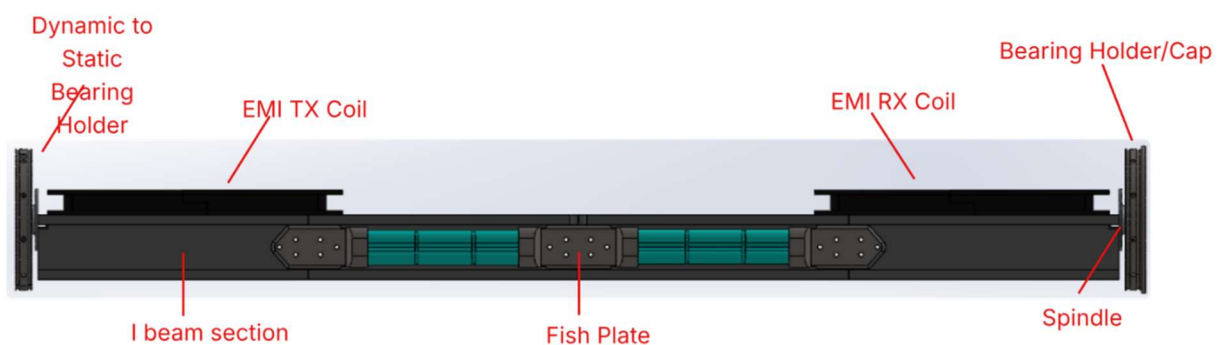


Figure 35 - Rotating Assembly Model

The beam is bolted to spindles at either end, manufactured from stainless steel billets on a CNC lathe to achieve a transition fit with the bearings that allow the beam to rotate. A transition fit was specified to ensure minimal play between the spindle and the bearing, thereby preventing oscillation and reducing fatigue to the bearing races and spindles themselves. To satisfy DR6, stainless steel sealed bearings were selected due to their lower conductivity compared to commonly available chrome steel bearings. The bearings were also selected such that the inner diameter was sufficient to allow slip rings and electrical connectors to pass through the interface between the static and dynamic sections of the buoy.

Two bearing holders were designed for the ends of the beam. As depicted on the right-hand side of figures 35 and 36, the bearing holder on this side is responsible for both securely mounting the bearing and sealing the end of the outer tube to prevent water ingress. It consists of a main body and a retaining plate bolted over the bearing with an interference fit, ensuring the bearing cannot slip within the housing. On the left-hand side in Figures 35 and 36, the dynamic to static bearing holder was designed as an asymmetric keyed component, allowing it to be manufactured as two identical printed halves with the bearing placed between them before assembling with stainless steel M5 hardware.

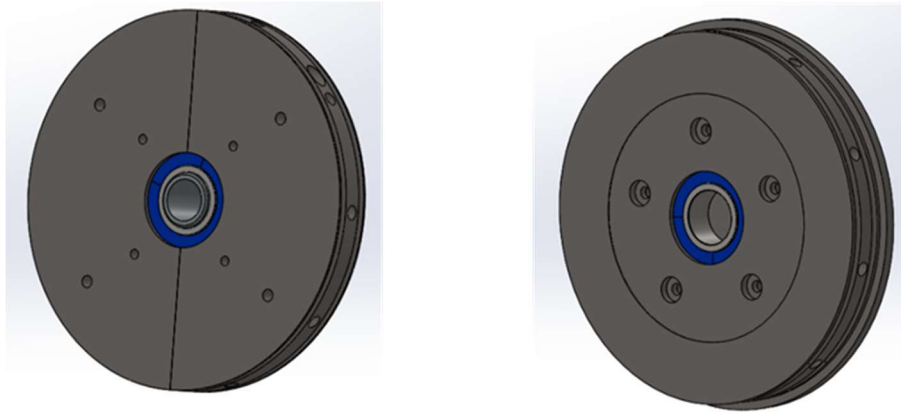


Figure 36 - Bearing holders, left: dynamic-static connector, right: dynamic side end-cap

By employing this novel internal rotor design, the EMI coils are able to passively self-orientate, thereby satisfying DR2, the require that maintain a HCP layout. Additionally, the orientation of sensors such as the cameras and UV sensor is maintained optimal throughout deployment, ensuring DR4 is met.

10.3.2. Static End

As the buoy requires the turbidity, salinity, and external temperature sensors to be mounted through the outer wall of the buoy, The entire internal structure could not rotate. Instead, a small section was designated to remain static relative and not rotating within the buoys body to accommodate these constraints. The designed sub-assembly for the static section can be seen below in Figure 37.

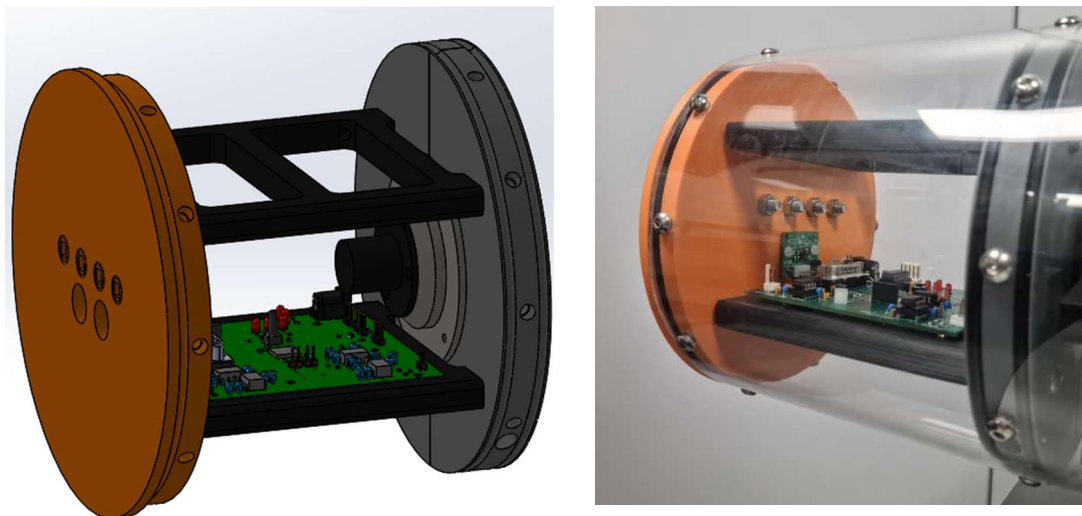


Figure 37 - Static end 3D render and prototype assembly

As shown in Figure 37, the static section consists of a structural PCB mounting plate for the daughter PCB and a support plate connected to the end of the buoy. The external end cap, shown in orange, includes both mounting holes and through holes for the salinity sensor probes and the turbidity sensor developed during the project. The temperature sensor is mounted through the tube wall within this section and connected directly to the daughter PCB. Additional consideration was given to sealing, with O-ring recesses incorporated such that an O-ring can be seated between the external caps face and the coach bolt flange to create a watertight seal. The static section also includes a mount for a 12-wire slip ring used to allow power and digital communication to transfer between the two sections of the buoy.

10.3.3. Electronics and Peripheral Mountings

The buoy's electronics and peripherals are mounted within the center of the rotating I-beam. To accommodate these components, a multi-level architecture was adopted to maximise the available internal volume of the tube. This can be seen in Figure 38, along with the arrangement of the components.

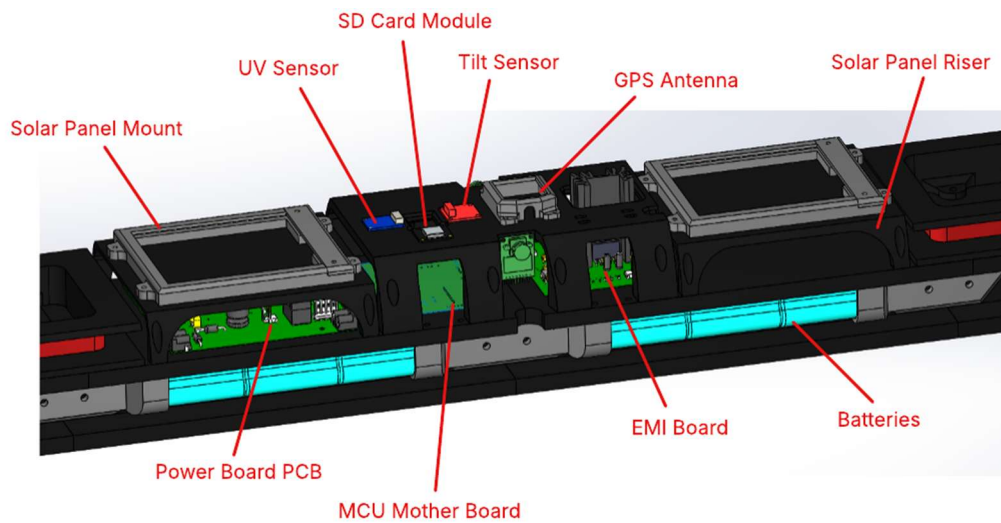


Figure 38 - Mounting of electronics and peripherals on the I-beam centre

As shown, the design elevates the solar panels and most sensors above the PCBs developed during the project. This approach not only provides the required clearance but also allows individual sensors to be replaced without full disassembly of the system, thereby meeting DR3. Particular care was taken in the placement of the tilt sensor to ensure it is aligned with the buoy's axis of rotation. Additionally, the UV sensor and solar array were positioned centrally to minimize attenuation and diffraction effects through the tube. The batteries are located below the main deck of the I-beam to provide a lower centre of mass, assisting the beam's self-levelling behaviour. The raised sections follow the same curvature of the tube with the same tolerance as the allowed for the I-beam to prevent the rotor striking the tubes internal wall.

10.3.4. Outer shell

The buoy's outer shell is a critical component, forming both the external casing and providing a structural mounting for the bearing holders. Discussions were held with manufacturers to assess the feasibility of producing a custom tube to NATO class B standard; However it was determined that while manufacture is possible, the cost and required unit quantity made the concept unfeasible [112]. Therefore a decision was made to use a 180 mm outer-diameter (OD), off the shelf, extruded acrylic tube. Although this is 5.3 mm larger in diameter than the NATO standard, no internal dimensions of the buoy were modified to benefit performance, particularly the I-beam and EMI coil width constraint. Consequently, the internal structure was designed as if a 174.7 mm diameter tube had been used, in accordance with the agreed specification.

10.4. Mechanical Test Assembly

Following completion of the final CAD-simulated assembly, part files were sent for manufacture and 3D printing, and a full physical test assembly of the mechanical system, without electronics, was completed. Figure 39 shows the physical test assembly.

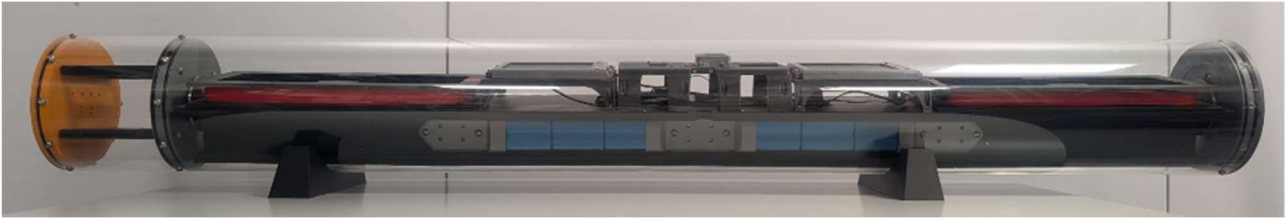


Figure 39 - Physical mechanical assembly

The assembly required 30 holes to be drilled through the tube at predefined locations to mount the bearing holders and end caps, along with four holes drilled and filed square for the temperature probe bolts. To ensure their accuracy a custom drill guide was fabricated to support this process. In accordance with DR6, all hardware used was stainless steel.

11. Full System Implementation

11.1. Electronic integration

As previously discussed the buoys electronics are divided across two main physical sections: the dynamic side and the static side. This separation was required because several subsystems depend on their position within the buoy. The dynamic side contains sensors and modules associated with the rotating section, including the EMI system, UV sensor, GPS, and Bluetooth module. In contrast, the static side contains sensors that interact directly with the water or are required to pass through the buoy's outer shell, including the turbidity sensor, salinity sensor, and IMU. To support this architecture, two custom STM32-based PCBs were developed; a Motherboard and a Daughter board, shown as label 'N' and 'M' respectively in Figure 2.

The motherboard is located on the dynamic side of the buoy and is built around a STM32F405RGT6 microcontroller. This PCB acts as the central control unit for the whole buoy, managing power distribution from the power board, program execution, sensor requests, serial communication, data storage, and communication with the Daughter board. It also includes dedicated headers for the dynamic-side sensors, allowing each module to be connected or disconnected easily before deployment. This modular approach improves survivability, simplifies fault finding, and provides cleaner cable management, as shown in Figure 40.

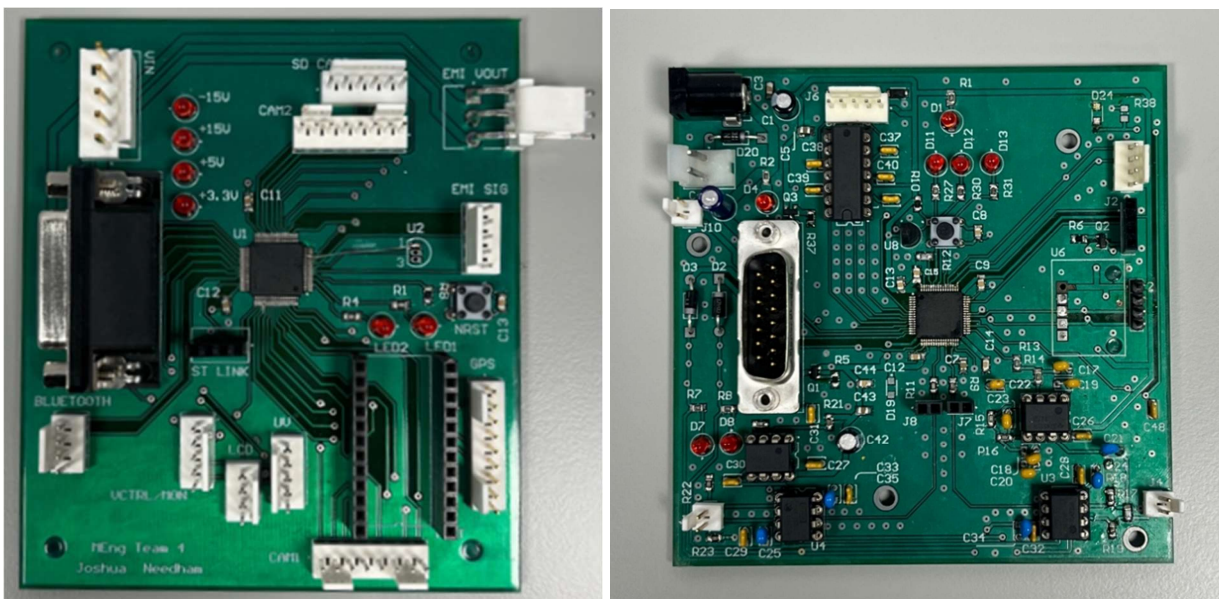


Figure 40 – Mother MCU Board (LHS) and Daughter Board (RHS)

The Daughter board as shown in Figure 40 is located on the static side of the buoy and is responsible for interfacing with the salinity, turbidity, the IMU and the external temperature sensors. It uses an STM32F411RET6 microcontroller, which was selected as a lower-power alternative to the Mother board microcontroller because the static-side sensors only need to be sampled periodically. The Daughter board also contains key analogue circuitry for the salinity subsystem, including the Sallen-Key filtering stage discussed in Section 5.3.

Communication between the Mother and Daughter boards is achieved through the slip ring using an asynchronous half-duplex serial link. Since communication is controlled from the Mother board, the Mother sends a request to the Daughter board and then waits for either the requested sensor data or a timeout response. This approach reduces unnecessary activity on the Daughter board, allowing it to remain in a low-power sleep mode until data is required. A structured message format is used for communication, as described in Section 8.3, containing a header to identify the response type and a CRC field for error checking. No analogue signal is passed through the slip-ring instead only power and digital communication, by digitising sensors readings on the Daughter Board it allows sensors reading to be transmitted without the risk of voltage drop or noise being introduced by the slip ring.

Overall, this two-board architecture provides a practical balance between modularity, power efficiency, and system reliability. By separating the electronics according to the physical layout of the buoy, sensor wiring is simplified, analogue circuitry remains close to the relevant sensors, and the Mother board can coordinate the complete system while the Daughter board response and performs low-power local sensor acquisition.

11.2. Firmware integration

The buoy firmware is split across two programs running on the Mother and Daughter boards. The Mother board runs the main program and controls the overall system timing, including different sampling schedules depending on whether the buoy is on land, ice, or sea. It requests sensor data from the Daughter board, logs the received measurements, and manages data transmission. Additionally, the system allows the daughter board to be put into a sleep state to conserve power and can be woken by the mother board.

The firmware integration of the sensor code was done using classes in C++ to encapsulate the sensors functions, not only providing a level of abstract useful for masking back-end functions but additionally allowing seamless integration as all that was required was to import the sensors class and make an instance of it.

11.3. Additional Sensors

Three additional sensors were integrated into the buoy platform to further extend its environmental monitoring capability. A commercial off-the-shelf LTR-390UV-01 UV sensor, shown as label P in Figure 2, was selected and programmed to retrieve both the raw UVS count and the Ambient Light Sensing (ALS) reading, from which the UV Index (UVI) was derived [113]. Monitoring UV irradiance is a valuable capability in polar regions, where ozone depletion events can cause record-breaking increases in surface UV radiation with significant consequences for polar ecosystems [114]. The ALS reading additionally provides a measure of ambient light levels at the buoy's location, which can be used to infer whether the buoy is submerged, ice-covered, or exposed to open sky, offering useful contextual information about the buoy's operational state.

For positioning, a GPS module, shown as label S in Figure 2, was integrated with a checksum algorithm applied to all incoming NMEA 0183 sentences, using an 8-bit XOR of sentence characters to detect and discard any corrupted transmissions before processing [115]. The NMEA output was parsed and split into its constituent fields, allowing latitude, longitude, altitude, and timestamp to be logged and transmitted independently. A water ingress detection circuit was also implemented using an 0805 capacitor pad connected to both a GPIO and ADC pin; the presence of water bridging the pad alters the measured response, providing an early warning of an enclosure breach.

Finally, an internal temperature sensor was incorporated onto the central motherboard, designated label N in Figure 2, to monitor the thermal environment within the buoy enclosure, ensuring internal conditions remain within the safe operating range of the onboard electronics, a critical consideration for the onboard electronics, notably the power system, given the extreme external temperatures expected during polar deployment.

11.4. Full System Assembly

Following the testing and integration of the subsystems on development boards the subsystem and firmware were transitioned to the Mother and Daughter PCBs and assembled in the buoy, Figure 41 below shows this completed assembly with the ground station running on the laptop to receive sensor data.



Figure 41 - Integrated buoy system and corresponding GUI ground-station

12. Project Summary

12.1. Introduction

Over the course of the project, the team successfully delivered a prototype climate-monitoring sensor buoy comprising of an EMI sea-ice thickness measurement system, a suite of environmental sensors, custom microcontroller hardware, a solar energy harvesting power architecture, and a GUI for remote data visualisation and control. Collaboration and task management remained strong throughout both semesters, and the final system satisfies the technical specifications defined in Section 1.5.

12.2. Achievements

Tables 12 and 13 below evaluate the degree to which the project objectives and specifications, outlined in Sections 1.4 and 1.5, were successfully achieved.

Table 12 - Technical achievement against objectives

Objective Number	Completion Status	Notes	Report Section
1	Completed	A comprehensive literature review of existing climate-monitoring buoys and sea-ice thickness measurement techniques was conducted, identifying the capabilities and limitations of current solutions to inform the system specification and design decisions.	1.6
2	Completed	A camera-based imaging system was designed and implemented, incorporating image processing algorithms that assess and filter image quality prior to storage and transmission. The system autonomously discards images not containing useful information, such as an obscured camera.	6

3	Completed	A modular suite of environmental sensors was successfully designed, integrated, and tested. Sensors include turbidity, salinity, UV irradiance, IMU, and external temperature, each validated through laboratory testing to confirm operation.	4, 5, 7, 9, 11
4	Completed	Custom microcontroller hardware was successfully developed, comprising a central motherboard and an auxiliary daughter board that together interface with and acquire data from all onboard sensing subsystems.	11.2
5	Completed	A cold-resilient, energy-harvesting power architecture was developed and tested, incorporating solar energy harvesting, battery management, and ultra-low quiescent sleep states to balance high-current sensing demands with extended autonomous operation.	3
6	Completed	A full-scale prototype buoy enclosure was designed and constructed, with all hardware subsystems successfully mounted within the assembly. The enclosure dimensions conform to the project specification and provides sufficient protection for all internal electronics.	10
7	Completed	All developed hardware subsystems were integrated into a unified prototype capable of simultaneous operation, forming a deployment-ready platform that brings together the EMI system, environmental sensors, power architecture, communications, and mechanical enclosure.	11
8	Completed	A low data rate communication system was implemented using BLE, enabling remote sensor data transmission and device configuration representative of satellite communication bandwidth constraints.	8
9	Completed	A GUI was developed that allows a user to request, organise, and visualise data from all onboard sensors. Sensor readings are displayed clearly and intuitively, supporting straightforward interpretation of environmental measurements.	8
10	In Progress	A fully integrated prototype has been assembled, with all subsystems physically mounted within the mechanical enclosure and operating simultaneously as a unified platform. All individual subsystem firmware has been independently developed, verified, and validated. Remaining work focuses on end-to-end system-level testing and refinement of cross-subsystem interactions to optimise overall platform performance as well as improving the GUI ground station.	11.4

Table 13 - Technical achievements against specifications

Specification Number	Completion Status	Notes	Report Section
1	In Progress	A non-destructive frequency domain resonant EMI system was presented for sea ice thickness measurement, making use of a bucking coil and a discrete lock amplifier to separate in phase and quadrature components. The system includes additional phasor rotation and subtraction to fully null the primary field. The system needs further development to optimise the bucking coil geometry and the opportunity to collect more data such that ice thickness can be accurately determined.	2

2	Completed	The buoy accurately measures a range of ECVs including sea-ice thickness, water turbidity, salinity, surface temperature, UV irradiance, and platform orientation, providing a comprehensive environmental monitoring capability.	4, 5, 7, 9, 11
3	Completed	A solar energy harvesting and battery management system was developed and tested, designed to sustain continuous sensing and communication loads over an extended autonomous deployment period.	4
4	Completed	The buoy mechanical enclosure and all internal hardware were designed and built to conform to the dimensional constraints of the project specification, with the overall system successfully housed within the required constraints.	4
5	In Progress	Most components meet the –40 °C rating, with the core exceptions being the batteries and mechanical construction materials, both held back by budget rather than availability. Replacements rated to the full specification have been identified for future iterations of the buoy, so the limitation is primarily related to cost rather than design choices.	1.5

Full end-to-end system integration has been substantially achieved. A complete, operational prototype has been assembled, with all hardware subsystems physically integrated within the mechanical enclosure and all firmware running simultaneously as a unified platform. Each subsystem has been independently developed, tested, and validated against its design requirements. Ongoing work is focused on system-level refinement, optimising cross-subsystem interactions, communication scheduling, and sensor data throughput, to maximise the robustness and performance of the platform.

12.3. Project Management

At the outset of the project, key responsibilities were assigned to ensure structured coordination and effective delivery. A project manager was appointed to define direction, monitor progress and manage task and resource allocation. An auditor oversaw budgeting and procurement activities to ensure expenditure aligned with project constraints. And a secretary-maintained meeting documentation, including minutes and agendas, and supported coordination of meeting schedules across team members and stakeholders.

12.3.1. Technical Objectives and planning

The project had overall objectives that were agreed at the project's conception, these objectives were decomposed into small tasks to be worked on over shorter time periods. These were allocated to individuals or in the case of larger and related work packages to specialised sub teams. The goals of each individual or team were agreed on by the wider project team to ensure clear expectations and a fair work distribution.

The team used a scrum meeting structure and therefore the progress on goals and objectives were reviewed continually. There was an expectation that team members would further decompose larger objectives into measurable goals achievable within a weeks' time, if these weekly objectives were not met, an explanation and actionable plan was required to be presented in the scrums highlighting the blockers causing the delay. As a result, when issues were raised the team could collectively discuss the problem, offer solutions and allocate resources appropriately in dynamic manner.

Towards the conclusion of the project when integration work was at its peak, scrum sessions were held twice weekly as to allow tasks to be managed more closely and feedback given more frequently.

12.3.2. Meetings

The team held at minimum twice weekly formal meetings, one independent of supervisors using the scrum structure and one with them. However, the majority of weeks the team held 2 meetings in addition to the supervisors meeting. At these meetings all members were required to attend in person by default, however under exceptional circumstances members and supervisors joined online. In addition to these formal meetings there were countless informal meetings and working session with team members both in subgroups and in general within the lab space.

Scheduling meetings was the responsibility of the project secretary who together with a team scheduling app was responsible for finding appropriate times that worked for all team members, this was critical as 3 team members held part-time employment and therefore had changing and at times tight schedules.

A weekly meeting agenda was shared with supervisors prior to their meeting, highlight key areas for discussion and focus. Meeting minutes and agreed actions were distributed to the team members following each meeting. This built on the scrums to ensure accountability and clear communication with supervisor.

12.3.3. Budget

The team auditor managed the budget using a detailed spreadsheet which tracked individual component costs throughout the project. The project was allocated a budget of £1,650, with the total expenditure reaching £1,384.89 to date. At the start of the project, it was agreed that each member could purchase components under £20 without needing team approval, helping speed up procurement. Most of the spending occurred in Semester 2, after the plan for individual subsystems were completed in Semester 1 and the development and testing procedures were finalised. Costs were primarily associated with the development of power subsystem and the buoy's mechanical design, while part of the initial contingency budget was used to replace components from earlier prototypes. Of the remaining £265.11, £150 has been reserved as contingency for potential microcontroller PCB remakes, with the remainder allocated towards building test equipment for the final demonstration.

12.3.4. Outreach

The team met with Dr Jeremy Wilkinson, Sea Ice Physicist, from the British Antarctic Survey to discuss the project and the teams proposed design. This meeting provided valuable insight into arctic conditions the realities of polar research equipments needs and current shortfalls. Overall, his experience and insight helped to confirm the choice in sensors equipped to the buoy and the mechanical design of the buoy.

12.3.5. Behaviour and Teamwork

When the group formed in September 2025 a team working agreement was written and signed by team members in addition to poster with the expectations written up and placed on the laboratory wall. This served as an agreed benchmark for members behaviour, agreeing simple conditions such as; meeting will take place in person by default, meetings will be arranged at least 24 hours in advance, team members agree to be accommodating to people's time and schedules both personal and work, to highlight a few. As result the team formed quickly and made a safe space for questions, feedback and technical success.

12.4. Changes from the original plan

12.4.1. Work Assignments

The breakdown of work was led by the project manager and agreed upon by the team. Alongside this, a Gantt chart was produced outlining the expected timeline of these work packages. Initially the team was split three ways in semester 1, to focus on what were deemed the most technically challenging subsystems; sea ice thickness measurement, image processing and communications. This plan was followed with the team delivering minimal viable products for these subsystems by the end of the first semester; however the team fell short of developing these subsystems to the level aimed for.

As a result, the original plan for semester 2 had to be adjusted to compensate for the need of these subsystems to have further development. However, the group responded well to the opportunity to revisit planning, now with more information of the scale of technical challenges and in the context of the wider project. Subsequently a choice was made to purchase an off the shelf UV sensor and power management board therefore reducing the workload required.

Overall, hardware development in semester 2 was done well with all sensors developed, tested and characterised on time and on schedule. Some sensor systems faced initial setbacks in procurement; however, team members pivoted to other tasks in these periods such as to keep pushing the project forward. Hardware timelines did get squeezed at points as the team faced several manufacturing issues with PCBs leading to delays, however working closely with the PCB facility the team were able to feedback board issues and prevent a repetition of faults.

Firmware was an area the team delivered well, both for the firmware in the buoy and the ground station. As a team, a compartmentalised structure was used, utilising classes to encapsulate firmware sections and aid integration. As a result, there were few dependencies allowing concurrent development of the code and a clear plan for integration.

Mechanical design was an area where the plan changed frequently in response to external delays and raised goals. Initially the team had planned to deliver the buoy's internal structure without the outer tube and mounts, however it was decided that a full prototype of the mechanical packaging would better demonstrate the innovative design taken. Before committing to this, the proposed resources, both time and financial, were presented to the team and supervisors for agreement. As mentioned, the mechanical build faced manufacturing delays and setbacks, in particular the CNC spindles were delayed and the original outer tube which cracked during manufacture. However enough contingency had been allowed and therefore a full system was still delivered on time and in specification.

In conclusion, the project demanded flexibility from the team at every stage, and at every stage the team delivered. Changes to work assignments were met with a positive response, using each reset as an opportunity to refocus and reprioritise, ultimately resulting in the full system presented in this report.

12.4.2. Technical Changes

The power subsystem, specifically the selection of the MPPT solar charge controller IC underwent several design iterations. The initial design was made around the Texas Instruments BQ24650. During the testing phase, it became clear that the initial IC required more complex supporting circuitry than anticipated, prompting us to look for alternatives. In addition to this, component delivery lead times caused significant delays to prototyping.

Another significant change was the switch from the STM32F401RE to the STM32F405RGT6 microcontroller. Initially, the STM32 Nucleo-F401RE was used in early testing for individual subsystems. However, the development of the EMI subsystem revealed that the F401RE lacked an internal DAC needed to generate the excitation signal for the coil. This led to a change in moving from a Nucleo development board to a custom PCB design.

12.4.3. Risk Management

At the project outset, expected risks of the project were determined and assessed based for the potential impact on development schedule. Component availability and long lead times were identified as a major project risk. To reduce the impact of delays, multiple orders were placed in advance for low-cost critical components such as the STM MCU's and supporting components. For components with long lead times, orders were placed before the final boards' designs were sign off to keep procurement aligned with the development schedule.

In addition, PCB fabrication delays also presented a risk, as design errors and faulty board could add a week or two every revision. To mitigate this, the team worked closely with the PCB facility to understand their workflow to understand how design decisions could affect fabrication time. Their manufacturing constraints which included minimum gap rules, clearances, rack sizing were all considered into board designs to mitigate avoidable delays. Multiple copies of each board revision were also ordered where possible, allowing development to continue even when boards were damaged during testing. This proved to be a good decision, as some fabricated boards contained poorly fitted pads that prevented further development. These measures helped ensure hardware delays had minimal impact on the overall project timeline.

12.5. Future Work

The current prototype successfully validates the core working principles of autonomous polar monitoring within a sonobuoy form factor. To transition from a laboratory-validated proof-of-concept to a field-deployable mission-ready platform, several key areas of future work have been identified across the mechanical, electrical, and software subsystems.

12.5.1. Mechanical Durability and Deployment

While the current chassis validates the internal rotating I-beam architecture and integration within sizing constraints, the transition to final materials is required before a sea trial can be conducted.

- Material Optimisation: For the external hull, future work must investigate a cold-rated, optically transparent polymer (e.g., impact-resistant Polycarbonate) to ensure sealing while maintaining unattenuated light transmission for the camera and solar arrays. Additionally, the internal parts must move away from 3D printed components at risk of delamination and explore the possibility of fibre reinforced 3D printed components in low temperature applications or homogenous materials.
- Deployment Logistics: Although out of scope for this phase, future work must address the air deployment system of the buoy. This involves integrating a parachute deployment mechanism and impact-absorption structures required for high-altitude airdrops, ensuring the internal self-levelling mechanism and sensors survive the impact.
- Environmental Validation: "Wet-testing" and pressure testing is required to validate the waterproofing, buoyancy, and long-term anti-fouling measures in a simulated polar environment.

12.5.2. Instrumentation and Sensing Refinement

Building on the successful sensing baseline, future work will focus on increasing data density and measurement precision.

- Depth Characterisation: The current temperature string utilises four sensors to estimate the water-to-air interface. Expanding this to a higher-density array, integrated with the IMU and tilt-sensor data, would allow the system to precisely calculate the buoy's submersion depth and dynamic orientation relative to the ice-seawater boundary.
- High-Resolution Imaging: To overcome the current RAM constraints of the STM32 microcontroller, a streaming pixel-processing approach, utilising a running-statistics algorithm such as Welford's method, could be implemented. This would allow key environmental metrics to be computed and images to be processed at significantly higher resolutions before being discarded, bypassing the need for full-frame storage.
- EMI Optimisation: The EMI system requires further characterisation data to map the I and Q response to sea water. To achieve this, a testing day is being arranged at a local outdoor brine pool, as this will provide a large and extensive body of water over which to conduct testing. Additionally, further refinement of the bucking coil geometry is scheduled in order to increase the primary field null prior to amplification. It is therefore expected that this will allow greater gain to be applied to

the RX coil and enable greater depth measurement capability and increased SNR. This future work is aimed to be completed within the upcoming work package.

- Wave Sensing Resolution: The current IMU implementation uses a 32-point FFT at a 10 Hz sample rate, yielding a frequency resolution of 0.3125 Hz per bin and a minimum detectable period of 3.2 seconds. Future work should investigate reducing the sample rate to 2 Hz with an increased FFT size, which would extend the detectable wave period range to cover long-period open-ocean swell.

12.5.3. Energy Autonomy and Power Scaling

The power system was designed for scalability, and several features are scheduled for full-scope integration.

- Thermal Management and Calibration: Future testing in environmental chambers is required to calibrate the battery monitoring and solar harvesting curves at true 0°C to -40°C extremes. This includes adjusting the resistor ratios on the MPPT controller to match the maximum power point voltage and investigating the integration of active heating elements to elevate battery temperatures to 0°C during the polar spring, thereby enabling charging cycles that are currently inhibited by the "cold-charge" safety logic.
- Discrete System Integration: The next hardware iteration will move from modular breakout boards to a fully integrated discrete PCB. This includes the implementation of the 5th power rail (5 V for Iridium), the full 2S12P battery architecture with a dedicated hardware BMS (e.g., TI BQ76905), and integrated eFuses for the high-current $\pm 15\text{V}$ rails.

12.5.4. Data Architecture and Predictive Modelling

The software ecosystem offers significant opportunities for data intelligence and remote scalability.

- Satellite Telemetry Integration: Transitioning the data link from local Bluetooth emulation to a global Iridium satellite transceiver is a primary objective. This involves integrating Short Burst Data (SBD) protocols into the existing firmware to support remote telemetry from high-latitude locations.
- Advanced Visualisation and Prediction: The GUI is designed for infinite scalability. Future versions will incorporate GPS geolocation mapping, real-time FFT analysis on IMU data, and 3D waveform visualisation for the EMI returns. Further improvements are aimed to be completed within the upcoming work package.
- Generative Prediction: Beyond real-time monitoring, integrating generative prediction modelling would allow the system to compare live metrics against historical climate data. This would enable the buoy to autonomously detect anomalous environmental shifts or predict ice-melt events based on current trajectory and historical BAS datasets.

13. Concluding Remarks

The buoy platform successfully demonstrates that a multi-variable climate-monitoring sensor buoy can be structurally and electrically integrated within a strict NATO Class B sonobuoy form factor to deliver high-fidelity, in-situ environmental data. By unifying non-destructive electromagnetic induction sea-ice thickness measurements with custom environmental sensing arrays, memory-efficient edge processing, and a cold-resilient, energy-harvesting power subsystem, this project presents a validated, end-to-end engineering solution. Developed in direct collaboration with the British Antarctic Survey, the successful physical realization of this autonomous prototype provides a reliable and scalable foundation for future mass air deployment. Ultimately, this technical contribution offers a viable, non-destructive pathway to close the critical polar observational gap, empowering researchers to improve predictive global climate modelling and advance our understanding of atmosphere-ice-ocean interactions in the world's most rapidly changing environments.

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